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Design of new didactic devices for teaching of control engineering

Rafaël Hueso Adelantado

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Prof. Michel Kinnaert

The co-supervision of
Laurent Catoire

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Rafaël Hueso Adelantado

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Abstract

This thesis addresses the design, modelling, identification and control of a ball-and-hoop apparatus in double configuration, intended for the teaching of control engineering. The apparatus extends the single-hoop system introduced by Wellstead and takes as its starting point the double-hoop project of Gurtner and Zemánek, in which the ball may leave a rolling surface, travel through the air and land on another.

A hybrid model of the system is derived. It gives the ball access to four modes, three rolling modes and a free-flight mode, obtained from the Euler-Lagrange formalism, together with the guards and reset maps connecting them. The fourth mode follows from the opening cut in the inner hoop, which the published model does not cover. The resulting hybrid automaton is implemented and verified on a set of open-loop scenarios.

A prototype is then designed, built and characterised. Three experiments are performed on it to determine its parameters. The agreement between the model and the measured trajectories gives a mean root mean square error of 2.47° on the ball position.

Four control tasks of increasing difficulty are then formalised in a common optimal control framework. Stabilisation and reference tracking are achieved on the bench with a linear quadratic regulator, and the latency of the vision chain is shown to determine the loop rate below which the closed loop diverges, independently of the choice of gains. The looping manoeuvre is planned by direct collocation and stabilised by a time-varying regulator. It succeeds in simulation but fails systematically on the prototype, and the cause is traced to the reduced-order model, in which the no-slip constraint makes a violation of the law of Coulomb invisible to the optimiser. The flying ball task is consequently studied in simulation only.

The apparatus is functional as a teaching platform, and the modelling assumptions that limit its use for the harder tasks are identified explicitly.

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List of Abbreviations

BLDC	Brushless Direct Current
CPR	Counts Per Revolution
FOC	Field-Oriented Control
FOV	Field Of View
LQG	Linear Quadratic Gaussian
LQR	Linear Quadratic Regulator
NBR	Acrylonitrile Butadiene Rubber
OCP	Optimal Control Problem
PID	Proportional-Integral-Derivative
PLA	Polylactic Acid
PMSM	Permanent Magnet Synchronous Machine
PPR	Pulses Per Revolution
RMSE	Root Mean Square Error
TVLQR	Time-Varying Linear Quadratic Regulator

1 Introduction

The objective of this master thesis is to design, control and implement a didactic device for the teaching of control engineering. The device selected in this document is a flying ball in a hoop. Students discover the dynamics of the system through lab sessions and learn to implement a control strategy bringing the ball onto the inner hoop of the prototype.

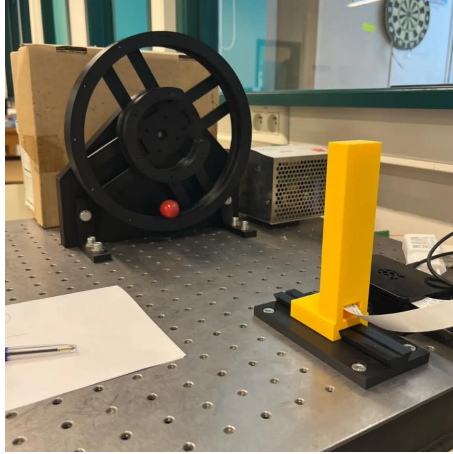


Figure 1.1: The prototype built in this work.

The ball and hoop is a well-known didactic device, with several commercial implementations available. It belongs to a family of classical benchmark systems used in control engineering education, alongside inverted pendulums and ball-and-beam apparatuses [1]. These systems share the important characteristic of being physically intuitive while still requiring careful attention when trying to model or control them.

In this system, the ball rolls inside a circular hoop, whose motion is controlled by a motor. The system studied in this thesis is further extended by a second hoop of smaller radius, mounted on the same axis. The dynamics become richer, since the ball can now leave a surface, travel through the air and land on the inner hoop. Such a system is no longer continuous, and is described as a hybrid system, alternating between continuous modes separated by discrete transitions [2]. The pedagogical value of the double-hoop configuration lies in its ability to demonstrate advanced control concepts, usually inaccessible during introductory courses, on a physically intuitive apparatus.

Four control tasks of increasing difficulty are formalised on this apparatus, from the stabilisation of the ball at an equilibrium to a manoeuvre in which it is launched, flies and is caught on the other hoop. Two of them are achieved on the prototype. The third fails on the bench while succeeding in simulation, and the analysis of that discrepancy is one of the results of this work.

This document begins with a state-of-the-art review, covering several ball-and-hoop prototypes and the strategies used to control them. The mathematical model of the complete system is then derived, including the equations of motion of each operating mode and the characterisation of the mode transitions. A chapter is dedicated to the choices made when designing the prototype, followed by the experimental identification of its parameters and the validation of the model against measurements. The control strategies are then implemented and their results reported, and a last chapter concludes with a summary of the achievements and perspectives for future work.

2 State Of The Art

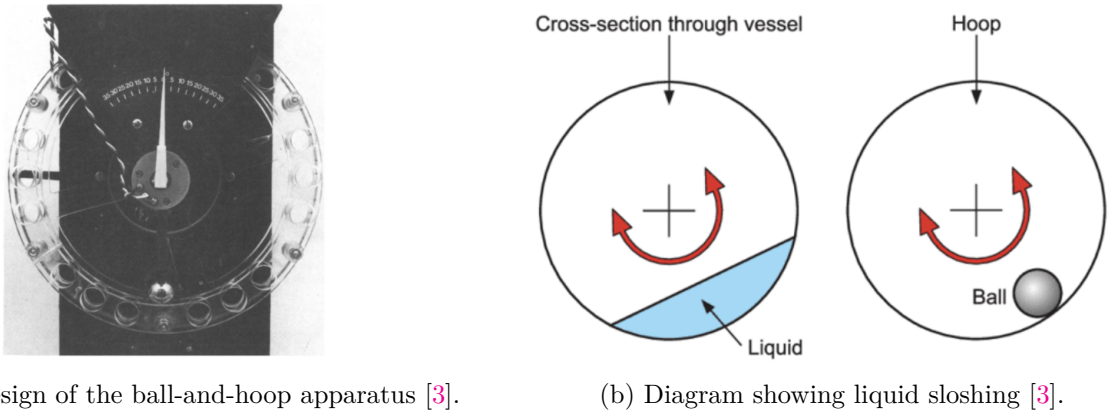
This chapter reviews the literature on ball-and-hoop systems, from the original single-hoop configurations to the more recent double-hoop extensions. Its purpose is to identify the properties that make this platform a versatile teaching tool, and to highlight the elements that motivate the present work.

The chapter starts by presenting the historical context and motivation behind the ball-and-hoop system, before discussing its key dynamical properties. A review of existing implementations is then provided, first for single-hoop configurations and then for the double-hoop extension. A short conclusion summarises the current state of the literature and positions this thesis with respect to the existing work.

2.1 Historical Context and Motivation

The ball-and-hoop system was introduced in the control engineering literature by Wellstead in 1983, in the configuration shown in [Figure 2.1\(a\)](#) [3]. The motivation was to provide control laboratories with a mechanical apparatus serving an educational purpose, on which students could observe and control a nontrivial dynamic behaviour.

The apparatus was also developed as a physical analogue of liquid sloshing in partially filled containers. Sloshing refers to the oscillatory motion of a liquid in a tank subjected to external accelerations, and can be approximated by the motion of a pendulum, which is itself analogous to the motion of a ball rolling inside a rotating hoop [3], [4]. This analogy is shown in [Figure 2.1\(b\)](#). The phenomenon is critical in spacecraft propellant management, where fuel slosh can destabilise the vehicle and its control [5], and appears in every domain involving liquid transport, from fuel tankers to aircraft storage and industrial pumping. The system therefore carries an industrial reading alongside its pedagogical one, which is part of what has kept it in use.



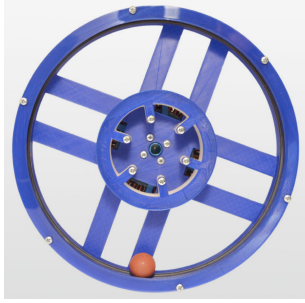
(a) First design of the ball-and-hoop apparatus [3].

(b) Diagram showing liquid sloshing [3].

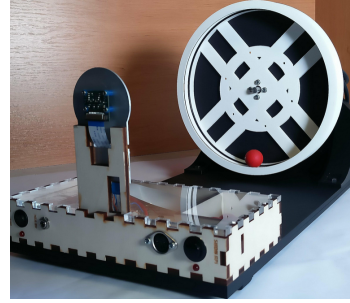
Figure 2.1: First ball-and-hoop apparatus, motivated by the analogy with liquid sloshing.

Following this publication, the system was commercialised and adopted in several university laboratories. Over the following decades, progressively more advanced control techniques were applied to the single-hoop configuration, from classical PID tuning [6] to Linear Quadratic Gaussian control [7], using both manual tuning and optimisation-based approaches [8], [9]. These studies confirm the versatility of this system as a control benchmark, covering a wide range of difficulty levels.

A significant extension was proposed later by Gurtner and Zemánek, who added a second hoop of smaller radius, mounted on the same axis [2]. The ball can then lose contact with a surface, travel through the air, and land on another. This transforms a purely continuous dynamical system into a hybrid one, and opens the apparatus to topics that the single-hoop configuration cannot reach, such as trajectory planning through mode transitions. The prototype is shown in Figure 2.2.



(a) Prototype.



(b) View from the ball-tracking camera.

Figure 2.2: Flying Ball in Hoop prototype developed by the AA4CC group [2].

Studies on double-hoop configurations nevertheless remain rare, and the extension is documented in essentially one peer-reviewed publication. That work establishes the feasibility of the apparatus and demonstrates two complex tasks on it, and the present thesis takes it as its starting point.

2.2 Key Properties

Two properties give the ball-and-hoop system its interest and its difficulty. The first is shared with a family of classical control benchmarks, and the second is what sets the double-hoop configuration apart from all of them.

2.2.1 Underactuation

The ball-and-hoop system belongs to the class of underactuated mechanical systems, defined as systems possessing fewer independent control inputs than degrees of freedom [10]. A single motor torque acts on the hoop, and the ball follows through the rolling contact alone. The two coordinates cannot be commanded independently, and the ball is never actuated directly.

Several classical benchmarks share this property. Figure 2.3 places the ball-and-hoop system among some of them [1]. The inverted pendulum, surveyed in over 140 references in a dedicated review [11], features an unstable upright equilibrium requiring active stabilisation. The reaction wheel pendulum [12] uses a rotational actuator to influence the motion of the pendulum. The ball-and-beam system [13] involves a ball rolling on a tilted beam, with simpler one-dimensional dynamics.

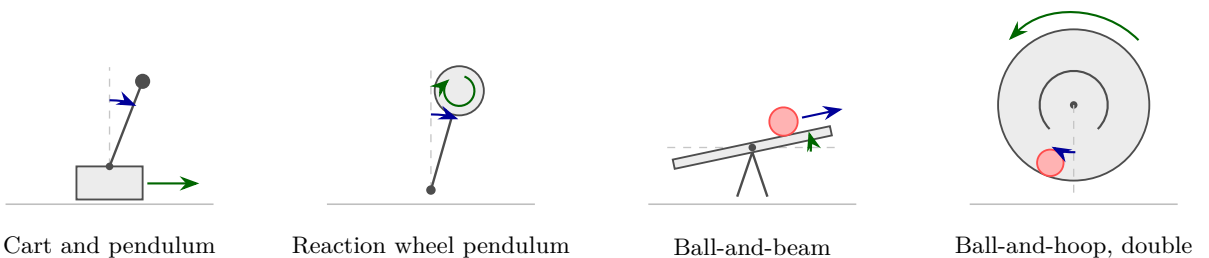


Figure 2.3: Common underactuated benchmarks. Each system has two degrees of freedom for a single input. The actuated coordinate is shown in green, the free one in blue.

All of them share the same count of two degrees of freedom for one input. They are distinguished by what happens at the contact between the actuated body and the unactuated one. The ball-and-hoop, in its double-hoop configuration, is the only one where the contact can be broken. The ball-and-hoop system also admits more equilibria than the others: each rolling surface admits its own, and the double-hoop configuration has three (see [section 6](#)). A controller can be designed for each of them separately, and more interestingly, the ball may be brought from one equilibrium to another.

Underactuation also has a frequency-domain reading. In the ball-and-hoop system, the transfer function from motor torque to ball position has a relative degree of four, since two integrations relate the torque to the hoop angle and two more relate the hoop motion to the ball position [3]. The two pendulums of [Figure 2.3](#) reach only two, the actuator acting on the same body whose motion is to be controlled, while the ball-and-beam reaches four for the same reason as the ball-and-hoop, the actuator tilting a beam that in turn drives the ball. This count assumes the torque to be the input. Commanding the angular acceleration of the hoop directly, as done in [section 6](#), removes the two integrations of the actuated body and brings the relative degree down to two.

A system of high relative degree responds to any input change with a phase lag that can be large, leaving little margin available for a feedback loop. Any additional lag, from a sensor, a filter or a computation for example, quickly consumes the phase that remains.

2.2.2 Hybrid Dynamics

A hybrid dynamical system evolves continuously within a given operating regime and undergoes abrupt transitions when certain conditions are met. The formal framework is built around three elements [14], [15]:

- a finite set of discrete modes, each with its own continuous dynamics,
- a set of guard conditions defining when a transition occurs,
- a set of reset maps specifying how the state is transformed at the transition instant.

The complete state is the pair formed by the continuous state and the active mode.

Such systems appear in many engineering applications, from walking robots alternating between support phases to gear shifting and power converters. In the double-hoop apparatus, the framework captures the physical behaviour directly. The ball alternates between rolling phases, where it is constrained to a hoop surface, and free-flight phases, where it moves under gravity alone, and each transition corresponds to a physically meaningful event, either a loss of contact or an impact.

The single-hoop configuration exhibits none of this. The ball remains on its track throughout, and the system is continuous. Adding a second hoop is what introduces the discrete transitions, which makes this apparatus one of the few educational benchmarks exhibiting hybrid behaviour naturally rather than by construction. The formalism itself is presented during the mathematical modelling of the system (see [section 3](#)).

2.3 Single-Hoop Systems

2.3.1 Physical Description

The single-hoop configuration consists of one circular hoop of radius R_h mounted on a horizontal axis and driven by an electric motor. A solid ball of radius R_b and mass m_b rolls along the inner surface of the hoop under the combined effect of gravity and of the rotation of the hoop [3]. The state is described by the hoop angle θ , the ball angular position ψ measured from the downward vertical, and their derivatives, with the motor torque τ as input, as shown in Figure 2.4.

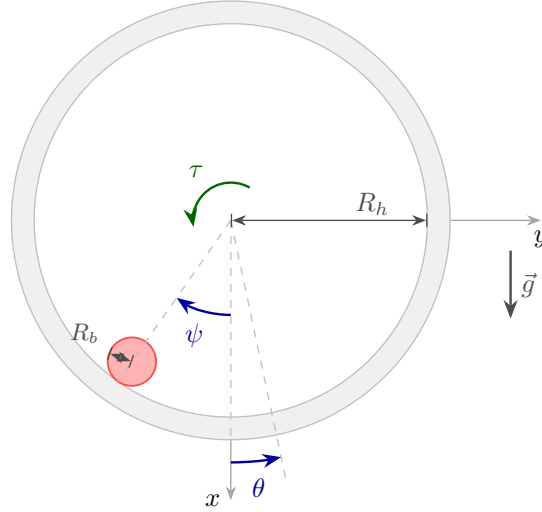


Figure 2.4: Single-hoop configuration and its coordinates.

The equations of motion are obtained by the Euler-Lagrange formalism, under the assumptions of pure rolling contact, rigid bodies and viscous friction at the contact and at the motor bearing. These equations are fully derived for the double-hoop configuration in the next section of this work (see [section 3](#)), of which the present configuration is a particular case. Two important features are commonly mentioned in the literature.

The first is the coupling between the ball and the hoop. Both ψ and θ appear in the equations. The hoop and the ball exchange angular momentum through the rolling contact, and neither can be moved without affecting the other. The second is the pendulum-like nature of the motion of the ball: the gravitational term is proportional to $\sin \psi$, so that for small angles the ball behaves as a damped harmonic oscillator, but departs from that behaviour as soon as the excursion grows.

Some authors simplify the model by assuming the motor to be powerful enough and fast enough for the hoop angle to be treated as the direct input. This simplification discards the motor equation and leaves a second-order relation between θ and ψ [3]. It removes the relative degree of four discussed in [subsection 2.2](#), and with it the difficulty that motivates most of the control work reported below. The present work keeps the cascade structure and the separation of time scales it relies on, but treats the commanded quantity as the hoop angular acceleration rather than the hoop angle, which preserves the dynamics of the hoop while acknowledging that the driver realises the command faster than the mechanical system responds. This choice is justified in [section 4](#) and verified on the hardware in [section 6](#).

2.3.2 Applications

The Ball and Hoop Apparatus CE109, developed by TecQuipment, is the commercial implementation of the system [16]. A steel ball rolls in a hoop of approximately 200 mm diameter driven by a DC servomotor, connected to a dedicated digital controller and to proprietary software for real-time implementation. Its purpose is pedagogical, and it supports a progressive sequence of experiments, from system identification through step and frequency responses to pole-zero analysis, PID design and cascade control with an inner velocity loop.

Several research groups have built custom prototypes to go beyond that sequence. Working outside a proprietary environment gives access to the low-level hardware, which is what allows optimal state feedback and automated tuning to be implemented and compared on the same plant [6], [7], [17]. These prototypes differ in their instrumentation, and in particular in how the ball position is measured, but they share the same dynamical structure and the same set of demonstrated tasks, namely stabilisation of the ball at the bottom of the hoop and tracking of a reference within a small neighbourhood of it.

2.3.3 Control Strategies

The single-hoop system has served as a basis for a broad range of control methods. Proportional-integral-derivative (PID) control is the first method used on this system [3]. Classical tuning rules give a workable starting point on this plant, but can produce underdamped responses, a direct consequence of the high relative degree of the system (see subsection 2.2). Better results are reported on research prototypes where manual tuning is replaced by optimisation algorithms, which search the best sets of gains automatically by simulating the closed loop and keeping the values that minimise a performance criterion [6], [8], [9], [18].

Cascade control addresses the same difficulty by splitting the loop into a fast inner loop on the hoop velocity and a slow outer loop on the ball position. This separation exploits the natural hierarchy of time scales of the system, and it is the architecture of the commercial apparatus [16].

State-space methods represent the system by its state variables rather than by an input-output relation. This allows a feedback action on all of them from the single available input. The linear quadratic regulator computes this feedback by minimising a cost combining a quadratic penalty on the state deviation and a quadratic penalty on the control effort. The relative weight of the two is the design freedom. The gains follow from the solution of a single algebraic equation rather than from an iterative adjustment, making the design reproducible, and the resulting closed loop carries stability margins that are guaranteed by construction, rather than obtained by trial [19]. All states must however be available, which is easily done for the ball and hoop positions, but not necessarily for their velocities. The linear quadratic Gaussian design removes this limitation by combining the regulator with a Kalman filter, which estimates velocities from position sensors [7].

2.4 Double-Hoop Systems

2.4.1 Physical Description

The double-hoop configuration adds a second hoop of smaller radius, mounted concentrically inside the first and rotating with it. The ball can still roll on the inner surface of the outer hoop, but it may now leave that surface, travel through the air, and land on the outer surface of the inner one. Holding the ball on top of the inner hoop amounts to balancing a disk on a disk [20].

Three modes therefore describe the motion [2]. The ball rolls on the outer hoop, the ball is in free fall with no surface acting on it, or the ball rolls on the inner hoop. Figure 2.5 shows how the system passes from one to another.



Figure 2.5: Modes of the hybrid description of the ball-and-hoop system and the transitions between them, following [2].

Each arrow is triggered by a guard condition, denoted γ , which is a scalar function of the state changing sign at the instant of the transition. Two of them are losses of contact, and two are impacts:

- γ_1 and γ_4 mark the moment when the hoop stops pushing on the ball, on the outer and on the inner surface respectively. The ball is held against the track only as long as the normal force acting on it counteracts gravity, and it leaves as soon as that is no longer the case.
- γ_2 and γ_3 mark the moment when a ball in free fall meets a surface again, the outer hoop in the first case and the inner one in the second.

Crossing a guard changes the mode. A loss of contact leaves the position and the angular velocity of the ball continuous, since no impulsive force acts at that instant, and only the radial variables are reinitialised. An impact does not, since the ball lands with a velocity that the surface cannot accommodate, and the state after the shock is prescribed by a reset map built from the spin the ball carried through its flight.

The prototype distributed with the reference project carries an opening in the inner hoop, visible on the drawings of the repository and in the demonstration videos [21], [22]. The opening gives the ball access to the other surface of this hoop. The published model does not cover this configuration.

One assumption underlies every rolling mode, namely that the ball rolls without slipping on the surface it touches. It relates the spin of the ball to the two other angular velocities and reduces the number of degrees of freedom accordingly. It holds only as long as the surface can supply the tangential force the motion demands. High-friction rails are used in reported prototypes [2], [23]. The ball is treated throughout the literature as touching a single point of a cylindrical surface. The geometry obtained once the rails are built departs from that idealisation, and the correction is introduced in section 5.

2.4.2 Applications

The reference implementation is the Flying Ball in Hoop project, developed by the Advanced Algorithms for Control and Communication group at the Czech Technical University in Prague, and shown in [Figure 2.2](#) [2]. It is the only peer-reviewed publication documenting a fully functional double-hoop system, and the drawings and codes are available in a public repository [21].

The hoops are 3D-printed, with rubber O-rings embedded in them serving as rails for a metal ball, and the assembly is driven by a brushless motor through a speed controller. The hoop angle needs no sensor, since it follows from integrating the input, and the ball position is measured by a camera whose images are processed on the same single-board computer that runs the control loop, both at 50 Hz. The angular velocity of the ball is estimated by an extended Kalman filter rather than measured, and the latency of the vision chain, of the order of two control periods, is compensated by predicting the state forward over that horizon. The dimensions of the prototype serve as a starting point for the present design and are reported in [section 4](#).

The camera is the one instrumentation choice that the double-hoop configuration forces. Contact-based sensors, such as the infrared arrays or the coupled encoders used on single-hoop prototypes, report nothing while the ball is airborne, which is precisely the phase the second hoop exists to produce. An observer reconstructing the ball position from the hoop dynamics fails for the same reason, since the rolling constraint that couples the two no longer holds. A contactless sensor observing the ball through both regimes is therefore the only viable option, at the cost of the acquisition and processing latency a camera introduces [17].

Two tasks are proposed on this prototype [22]:

- the loop-the-loop, in which the ball starts at rest at the bottom of the outer hoop, completes a full revolution while keeping contact, and returns to rest. It must move fast enough over the top for the track to keep pushing on it.
- the flying ball, in which the ball leaves the outer hoop, falls, and is caught and held on the inner one. The state at the moment of departure must be chosen so that the flight and the impact that follow leave the ball in a condition the regulator of the inner hoop can absorb.

Both tasks go beyond the stabilisation and small-signal tracking reported on single-hoop systems. The model published alongside them is stated at the level of detail they require. The dynamics of the hoop are not modelled, since its angular acceleration is assumed to be commanded directly. The ball is assumed to roll without slipping, and to keep a constant spin during free flight.

Neither task uses the opening in the article, since the ball is caught on the convex face of the inner hoop during the flying ball scenario developed in it. The present work performs the same two tasks on an apparatus that carries the opening, which brings in a fourth mode and the transitions that go with it (see [section 3](#)).

2.4.3 Control Strategies

The tasks performed on the double-hoop configuration are not reachable with the methods used on single-hoop configurations. All of them stabilise the system near an equilibrium, but a loop-the-loop requires the ball to travel far from any equilibrium, and a flying ball passes through a phase in which the ball cannot be acted upon at all. The strategy developed to solve these tasks separates the problem into two, computing a nominal trajectory offline and stabilising it online [2].

The trajectory is obtained by numerical optimal control. Rather than choosing a reference by hand and hoping it can be followed, this method searches for the input history that minimises a chosen cost while satisfying the dynamics of the system and its physical limits. The search is made by direct collocation, which replaces the continuous problem by a finite one where the trajectory is described by its values at several instants. The dynamics become algebraic conditions between neighbouring instants [24]. The obtained trajectory is nominal, and any mismatch can accumulate. A feedback layer is therefore added, obtained by linearising the system along that trajectory rather than at a fixed point. This two-step approach is what makes both tasks feasible.

2.5 Conclusion

The literature on ball-and-hoop systems is strongly asymmetric. Four decades of work cover the single-hoop configuration, from the original apparatus of Wellstead to commercial implementations and research prototypes, with control methods ranging from classical tuning to optimal state feedback. All of it addresses the same regime, in which the ball stays near the bottom of its track and the system remains continuous.

The double-hoop extension is documented in a single peer-reviewed publication. That work models the apparatus as a hybrid system, poses two tasks that no linear method can solve, and demonstrates one of them on hardware. It establishes that the extension is feasible and that it is worth the additional complexity, and it is the starting point of the present thesis rather than a gap to be filled.

Three elements are nevertheless left open by the existing work, and they define the scope of the chapters that follow:

- The published model describes three modes and stops there. The prototype distributed with the project carries an opening in the inner hoop, which gives the ball access to a fourth rolling configuration that the model does not cover.
- The second task, in which the ball leaves one hoop and is caught on the other, is validated in simulation only. Its behaviour on hardware remains unknown.
- No assembly procedure or acceptance criterion is reported, so that reproducing the apparatus requires the reader to rediscover the choices that make it work.

This thesis addresses these three points. [section 3](#) derives the four modes explicitly, together with the guards and reset maps connecting them. [section 4](#) and [subsection 4.4](#) describe the design and the assembly of a prototype, with the identification campaign needed to give the model its parameters. [section 6](#) formalises four tasks of increasing difficulty on that prototype, from stabilisation to the flying ball, and reports what each of them achieves in simulation and on the bench.

3 Mathematical Modelling

The previous chapter described the ball-and-hoop apparatus and the behaviour reported on it. This chapter derives the equations governing that behaviour, for every configuration the double-hoop system admits. The derivation follows the Lagrangian framework and takes as its starting point the model of Gurtner and Zemánek [2], which covers three modes. The apparatus of this work carries an opening in the inner hoop, which gives the ball access to a fourth configuration, so the model is extended accordingly.

The chapter is organised as follows. This first section sets the frame, the parameters and the general form in which every equation of motion is written. The rolling modes are then derived, followed by the free-flight mode, where the contact disappears altogether. The transitions connecting these four modes are treated next, and the hybrid system is assembled from them. The model is finally implemented numerically, validated against a set of open-loop scenarios, and its limitations are stated.

3.1 Modelling Framework

3.1.1 Frame and Coordinates

The system is analysed in a fixed inertial frame (O, x, y) , with the origin at the centre of rotation of the hoops. The x -axis points downward, in the direction of gravity, and the y -axis points horizontally to the right. The position of the ball is described in polar coordinates (r, ψ) , and the rotation of the hoops by the angle θ . All angles are measured from the downward vertical and are positive in the counter-clockwise direction. These conventions are shown in Figure 3.1.

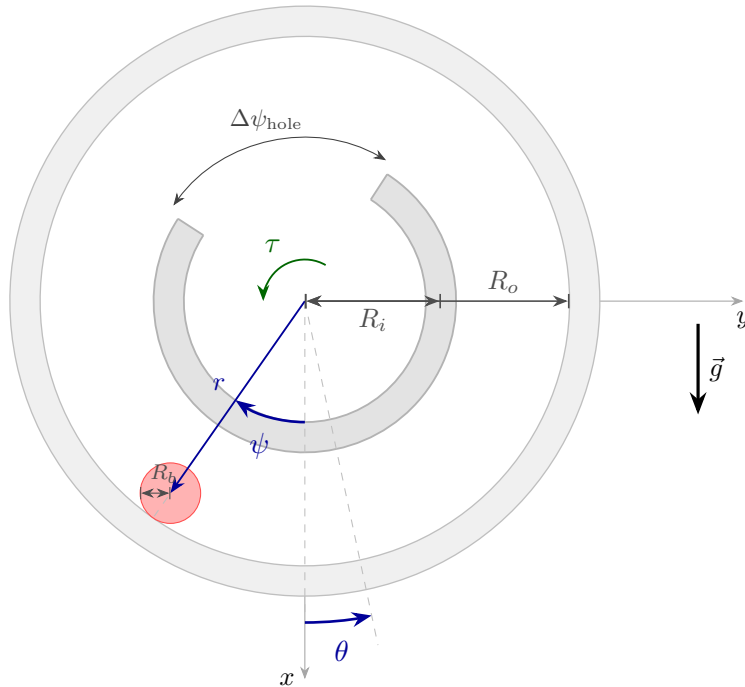


Figure 3.1: Diagram of the double-hoop system configuration.

The continuous state of the system gathers these coordinates and their derivatives,

$$\mathbf{x} = [r, \dot{r}, \theta, \dot{\theta}, \psi, \dot{\psi}]^T. \quad (3.1)$$

The radial distance r locates the centre of mass of the ball with respect to the rotation axis, and ψ its angular position in the fixed frame. During the rolling modes, the ball is held against a surface, r is constant and the system reduces to two degrees of freedom. During free flight, all three coordinates evolve freely.

A fourth variable is needed to describe the motion completely. The ball turns about its own centre, and that rotation, called the spin and denoted φ , carries part of the kinetic energy of the system. In the rolling modes the spin is not independent, since the contact relates it to θ and ψ . In free flight the relation no longer holds and the spin evolves on its own. Both cases are treated with the modes they belong to.

3.1.2 Parameters

The physical parameters appearing in the model are shown in [Figure 3.1](#) and collected in [Table 3.1](#), with their units. Their proper values follow from the components selection (see [section 4](#)) and are identified on the prototype itself (see [section 5](#)).

Symbol	Description	Unit
g	Gravitational acceleration	m/s ²
R_b	Ball radius	m
m_b	Ball mass	kg
I_b	Ball moment of inertia	kg·m ²
b_b	Ball rolling friction coefficient	N·m·s/rad
R_o	Outer hoop inner radius	m
R_i	Inner hoop mid-radius	m
t_i	Inner hoop wall thickness	m
$\Delta\psi_{\text{hole}}$	Angular width of the opening	rad
ψ_{hole}	Angular position of the opening centre	rad
I_h	Total moment of inertia of the rotating assembly	kg·m ²
b_m	Motor viscous friction coefficient	N·m·s/rad

Table 3.1: Physical parameters of the ball-and-hoop system.

The ball can roll on three surfaces. The inner surface of the outer hoop is concave, of radius R_o . The inner hoop, of mid-radius R_i and wall thickness t_i , presents a concave surface of radius $R_{i,\text{in}} = R_i - t_i/2$ and a convex one of radius $R_{i,\text{out}} = R_i + t_i/2$. The opening cut in its wall, of angular width $\Delta\psi_{\text{hole}}$ and centred at ψ_{hole} in the hoop frame, is what gives the ball access to the concave surface. On each surface, the centre of the ball travels along a circle whose radius depends on the curvature. The ball sits inside a concave surface and outside a convex one, so that

$$R_{\text{eff}} = \begin{cases} R_o - R_b & \text{on the outer hoop,} \\ R_{i,\text{in}} - R_b & \text{on the inner hoop, concave surface,} \\ R_{i,\text{out}} + R_b & \text{on the inner hoop, convex surface.} \end{cases} \quad (3.2)$$

This effective radius is the distance from the rotation axis to the centre of the ball whenever contact is maintained, and it replaces r in the rolling modes.

The ball is treated here as touching the surface at a single point. The tracks fitted to the prototype hold it on two rings instead, which displaces the centre of the ball and the radius at which it rolls. This correction is derived once the tracks are designed (see [section 5](#)).

3.1.3 Lagrangian Formulation

The equations of motion of each mode are derived using Lagrangian mechanics. The Lagrangian is defined as $\mathcal{L} = T - V$, where T is the kinetic energy of the system and V its potential energy. The generalised coordinates used in this formulation are $q = [r, \theta, \psi]^\top$, and for each of them the Euler-Lagrange equations read

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \quad (3.3)$$

where Q_i gathers the generalised forces associated with the coordinate q_i .

Friction is introduced through the Rayleigh dissipation function $\mathcal{D}$, a quadratic function of the velocities whose derivative gives the dissipative force on each coordinate. The equations then become

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial \mathcal{D}}{\partial \dot{q}_i} = Q_i^{\text{ext}}, \quad (3.4)$$

where Q_i^{ext} contains only the external non-dissipative forces, here the action of the motor. Writing the dissipation this way is one of two equivalent routes, the other being to express the friction forces directly as generalised forces Q_i in (3.3). Both give the same equations, and the dissipation function is retained here because it gathers every friction term in a single scalar [25].

Two assumptions are made at this stage and hold throughout the chapter. The ball and the hoops are treated as rigid bodies, which is justified by the stiffness of the materials with respect to the loads encountered in normal operation. The motion is also assumed to remain in the vertical plane containing the rotation axis. Out-of-plane motion is prevented by the mechanical design, the hoops rotating about a fixed horizontal axis and the ball being laterally constrained by the width of the tracks. The measures taken during assembly to enforce this second assumption are described in section 5.

For every mode, the resulting equations are written in the matrix form

$$\boxed{M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q) = B u} \quad (3.5)$$

where M is the mass-inertia matrix, C gathers the Coriolis, centrifugal and damping terms, G is the gravity vector, B is the input distribution vector and u is the input.

3.1.4 Input of the System

The quantity commanded in this work is the angular acceleration of the hoops, $u = \ddot{\theta}$. However, the motor delivers a torque. The driver uses its own control loop around the current (and therefore around the torque), at a rate much faster than the mechanical system responds. Seen from the outer loop, the commanded quantity is reached instantaneously, and the acceleration can be prescribed directly.

This cascade structure is the one adopted in the reference work [2]. The torque corresponding to a commanded acceleration follows from the equation of motion of the hoop, and is recovered whenever the actuator limits are checked.

With the frame, the parameters and the general form fixed, the equations of motion can be derived. The three rolling modes share enough structure to be treated together, and come first. Free flight, where the contact disappears and the two bodies separate, follows.

3.2 Rolling Modes

The three rolling surfaces that have been described lead to three different modes. They share the same global structure, since the ball is held on a circular surface turning about the same axis in each of them, but differ through the radius of the surface and the sign of its curvature. The derivation is therefore carried out once, in full, for the outer hoop, and the two remaining modes follow by substitution.

One assumption governs this derivation: the ball is assumed to roll on the contact point without slipping. Given the ratio between the radius of the ball and the ones of the surfaces, this assumption is reasonable.

3.2.1 Rolling Constraint

Rolling without slipping means that the two bodies in contact have the same velocity at the contact point. The point of the ball touching the surface is then momentarily at rest with respect to it, a condition that directly relates the spin of the ball with the two other angular velocities. When the ball rolls, for instance, on the inner surface of the outer hoop, the contact point moves with the tangential velocity at the contact radius, $v_{\text{hoop}} = R_o \dot{\theta}$.

The same point, seen from the ball, moves for two reasons: the centre of the ball travels along a circle of radius R_{eff} , and the ball turns about this centre. The tangential velocity of the contact point on the ball is therefore $v_{\text{ball}} = R_{\text{eff}} \dot{\psi} + R_b \dot{\varphi}$. Respecting the no-slip condition means that

$$R_o \dot{\theta} = R_{\text{eff}} \dot{\psi} + R_b \dot{\varphi}, \quad \text{hence} \quad \dot{\varphi} = \frac{R_o}{R_b} \dot{\theta} - \frac{R_{\text{eff}}}{R_b} \dot{\psi}. \quad (3.6)$$

The spin of the ball is thus determined by $\dot{\theta}$ and $\dot{\psi}$, and the system retains two degrees of freedom in this mode. The same reasoning applies to the concave surface of the inner hoop, with $R_{i,\text{in}}$ replacing R_o . On the convex surface, the geometry differs, and the contact point on the ball moves in the opposite direction relative to its centre. The condition becomes

$$\dot{\varphi} = -\frac{R_{i,\text{out}}}{R_b} \dot{\theta} + \frac{R_{\text{eff}}}{R_b} \dot{\psi}. \quad (3.7)$$

The three cases are gathered by introducing the radius R of the surface on which the ball rolls, and a sign σ carrying its curvature,

$$\dot{\varphi} = \sigma \left(\frac{R}{R_b} \dot{\theta} - \frac{R_{\text{eff}}}{R_b} \dot{\psi} \right), \quad \sigma = \begin{cases} +1 & \text{on a concave surface,} \\ -1 & \text{on a convex surface,} \end{cases} \quad (3.8)$$

with the values of R , R_{eff} and σ for each mode collected in [Table 3.2](#).

Mode	R	R_{eff}	σ
Rolling on the outer hoop	R_o	$R_o - R_b$	+1
Rolling on the inner hoop, concave surface	$R_{i,\text{in}}$	$R_{i,\text{in}} - R_b$	+1
Rolling on the inner hoop, convex surface	$R_{i,\text{out}}$	$R_{i,\text{out}} + R_b$	-1

Table 3.2: Parameters of the three rolling surfaces.

3.2.2 Derivation on the Outer Hoop

In this mode the ball is held against the inner surface of the outer hoop, so that $r = R_{\text{eff}}$ is constant and $\dot{r} = \ddot{r} = 0$. The generalised coordinates reduce to θ and ψ .

Kinetic energy

The kinetic energy gathers the motion of the ball and that of the hoops. The centre of mass of the ball travels along a circle of radius R_{eff} at the speed $R_{\text{eff}}|\dot{\psi}|$ and turns about its own centre, and the rolling constraint expresses this rotation in terms of the two generalised velocities (as seen in (3.8)). The hoops rotate about the fixed axis. The contribution of each of them gives

$$T_{\text{ball}} = T_{\text{ball,trans}} + T_{\text{ball,rot}} = \frac{1}{2} m_b R_{\text{eff}}^2 \dot{\psi}^2 + \frac{1}{2} I_b \dot{\varphi}^2 \quad (3.9)$$

$$T_{\text{hoop}} = \frac{1}{2} I_h \dot{\theta}^2, \quad (3.10)$$

and the total kinetic energy is the sum of these contributions,

$$T = \frac{1}{2} m_b R_{\text{eff}}^2 \dot{\psi}^2 + \frac{1}{2} I_b \left(\frac{R_o}{R_b} \dot{\theta} - \frac{R_{\text{eff}}}{R_b} \dot{\psi} \right)^2 + \frac{1}{2} I_h \dot{\theta}^2. \quad (3.11)$$

Potential energy

The only contribution comes from the ball, since the hoops turn about their own centre of mass. Since the x -axis points downward, the height of the centre of the ball above the origin is $-R_{\text{eff}} \cos \psi$, and taking the reference at the rotation axis,

$$V = -m_b g R_{\text{eff}} \cos \psi. \quad (3.12)$$

Dissipation

Energy is dissipated at the bearing of the motor and at the contact between the ball and the surface. Both of them are modelled as viscous, proportional to the corresponding velocity. This is an idealisation, as other dissipation sources may have an influence too: rolling resistance, static friction, aerodynamic drag [23]. Each of them is estimated for the present geometry in Appendix A, showing that drag is negligible, static friction stays within what the tracks can supply and that rolling resistance is not negligible but cannot be carried into a linear model.

The dissipation associated with the contact acts on the spin of the ball, leading to

$$\mathcal{D} = \frac{1}{2} b_m \dot{\theta}^2 + \frac{1}{2} b_b \dot{\varphi}^2 = \frac{1}{2} b_m \dot{\theta}^2 + \frac{1}{2} b_b \left(\frac{R_o}{R_b} \dot{\theta} - \frac{R_{\text{eff}}}{R_b} \dot{\psi} \right)^2. \quad (3.13)$$

Equation of motion for the hoop

The Lagrangian is $\mathcal{L} = T - V$. Differentiating with respect to $\dot{\theta}$ and collecting the terms gives

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = I_b \frac{R_o}{R_b} \left(\frac{R_o}{R_b} \dot{\theta} - \frac{R_{\text{eff}}}{R_b} \dot{\psi} \right) + I_h \dot{\theta} = \left(I_h + I_b \frac{R_o^2}{R_b^2} \right) \dot{\theta} - I_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\psi}, \quad (3.14)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \left(I_h + I_b \frac{R_o^2}{R_b^2} \right) \ddot{\theta} - I_b \frac{R_o R_{\text{eff}}}{R_b^2} \ddot{\psi}. \quad (3.15)$$

The Lagrangian does not depend on θ itself, so $\partial \mathcal{L} / \partial \theta = 0$. This reflects the rotational symmetry of the apparatus, the equilibria being independent of the absolute orientation of the hoops.

The dissipative contribution is

$$\frac{\partial \mathcal{D}}{\partial \dot{\theta}} = b_m \dot{\theta} + b_b \frac{R_o^2}{R_b^2} \dot{\theta} - b_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\psi}, \quad (3.16)$$

and the equation of motion for the hoop follows,

$$\left(I_h + I_b \frac{R_o^2}{R_b^2} \right) \ddot{\theta} - I_b \frac{R_o R_{\text{eff}}}{R_b^2} \ddot{\psi} + \left(b_m + b_b \frac{R_o^2}{R_b^2} \right) \dot{\theta} - b_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\psi} = \tau \quad (3.17)$$

Equation of motion for the ball

The same operations on ψ give

$$\frac{\partial \mathcal{L}}{\partial \dot{\psi}} = -I_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\theta} + \left(m_b R_{\text{eff}}^2 + I_b \frac{R_{\text{eff}}^2}{R_b^2} \right) \dot{\psi}, \quad (3.18)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = -I_b \frac{R_o R_{\text{eff}}}{R_b^2} \ddot{\theta} + \left(m_b R_{\text{eff}}^2 + I_b \frac{R_{\text{eff}}^2}{R_b^2} \right) \ddot{\psi}, \quad (3.19)$$

$$\frac{\partial \mathcal{L}}{\partial \psi} = -m_b g R_{\text{eff}} \sin \psi, \quad (3.20)$$

the last expression being the gravitational term, which is the only nonlinearity of the model. The dissipative contribution is

$$\frac{\partial \mathcal{D}}{\partial \dot{\psi}} = -b_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\theta} + b_b \frac{R_{\text{eff}}^2}{R_b^2} \dot{\psi}, \quad (3.21)$$

and the equation of motion for the ball follows,

$$\left(-I_b \frac{R_o R_{\text{eff}}}{R_b^2} \ddot{\theta} + \left(m_b R_{\text{eff}}^2 + I_b \frac{R_{\text{eff}}^2}{R_b^2} \right) \ddot{\psi} - b_b \frac{R_o R_{\text{eff}}}{R_b^2} \dot{\theta} + b_b \frac{R_{\text{eff}}^2}{R_b^2} \dot{\psi} + m_b g R_{\text{eff}} \sin \psi = 0 \right) \quad (3.22)$$

No input appears in this equation, which is the analytical expression of the underactuation discussed in the state of the art (see [subsection 2.2](#)). The motor acts on the hoop, and the ball responds only through the coupling terms.

3.2.3 General Form of the Rolling Modes

Writing $q = [r, \theta, \psi]^\top$ with $r = R_{\text{eff}}$ constant, the rolling modes take the form of (3.5) with

$$\begin{aligned} M &= \begin{pmatrix} m_b & 0 & 0 \\ 0 & I_h + I_b \frac{R^2}{R_b^2} & -I_b \frac{R R_{\text{eff}}}{R_b^2} \\ 0 & -I_b \frac{R R_{\text{eff}}}{R_b^2} & m_b R_{\text{eff}}^2 + I_b \frac{R_{\text{eff}}^2}{R_b^2} \end{pmatrix}, & G &= \begin{pmatrix} 0 \\ 0 \\ m_b g R_{\text{eff}} \sin \psi \end{pmatrix}, \\ C &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & b_m + b_b \frac{R^2}{R_b^2} & -b_b \frac{R R_{\text{eff}}}{R_b^2} \\ 0 & -b_b \frac{R R_{\text{eff}}}{R_b^2} & b_b \frac{R_{\text{eff}}^2}{R_b^2} \end{pmatrix}, & B &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \end{aligned} \quad (3.23)$$

In this matrix form, the radii values are written using the general notations R and R_{eff} . This is because the derivation is the same on the two other rolling surfaces.

The structure of (3.23) shows where the difficulty of the system lies: The matrices M and C have non-zero off-diagonal terms, so that the two coordinates cannot be treated separately. The gravity vector acts on the ball alone, and depends on ψ through a nonlinearity. The input vector acts on the hoop alone, so that the ball is reached only through the coupling. These three observations influence the whole control design (see section 6).

3.3 Free-Flight Mode

When the ball leaves a surface, the contact disappears and with it the constraint that tied the radial coordinate to the geometry. The three generalised coordinates become free, and the motion of the ball separates completely from that of the hoops.

3.3.1 Equations of Motion

Energies and dissipation

The centre of the ball now moves with a radial and a tangential component, while the hoops keep turning under the action of the motor,

$$T_{\text{ball}} = \frac{1}{2} m_b \left(\dot{r}^2 + r^2 \dot{\psi}^2 \right), \quad (3.24)$$

$$T_{\text{hoop}} = \frac{1}{2} I_h \dot{\theta}^2. \quad (3.25)$$

The potential energy retains the same form as before, with r in place of R_{eff} , and the dissipation function reduces to the bearing of the motor, since no contact remains,

$$V = -m_b g r \cos \psi, \quad \mathcal{D} = \frac{1}{2} b_m \dot{\theta}^2. \quad (3.26)$$

Aerodynamic drag on the ball is neglected here as well, its magnitude being small compared with gravity and inertia at the velocities and over the durations encountered on this apparatus (see Appendix A).

Hoops

The Lagrangian depends on θ only through its derivative, and the equation of motion for the hoops is that of a rotating body subject to viscous friction,

$$\boxed{I_h \ddot{\theta} + b_m \dot{\theta} = \tau} \quad (3.27)$$

The ball does not appear in it, and the hoops do not appear in the equations of the ball either.

Ball

Applying the Euler-Lagrange equations to the radial and angular coordinates gives

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{r}} = m_b \ddot{r}, \quad \frac{\partial \mathcal{L}}{\partial r} = m_b r \dot{\psi}^2 + m_b g \cos \psi, \quad (3.28)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = 2m_b r \dot{r} \dot{\psi} + m_b r^2 \ddot{\psi}, \quad \frac{\partial \mathcal{L}}{\partial \psi} = -m_b g r \sin \psi. \quad (3.29)$$

Dividing by m_b and rearranging gives the equations of motion of the ball, which match the expressions reported in [2].

$$\begin{cases} \ddot{r} = r\dot{\psi}^2 + g \cos \psi \\ \ddot{\psi} = -\frac{1}{r} (g \sin \psi + 2\dot{r}\dot{\psi}) \end{cases} \quad (3.30)$$

These two equations describe a projectile under uniform gravity. In Cartesian coordinates, the trajectory is exactly parabolic, and (3.30) is the same equation written in the polar frame. The term $r\dot{\psi}^2$ is the centrifugal contribution to the radial equation, while $2\dot{r}\dot{\psi}$ the Coriolis contribution to the angular one.

Matrix form

$$\begin{aligned} M &= \begin{pmatrix} m_b & 0 & 0 \\ 0 & I_h & 0 \\ 0 & 0 & m_b r^2 \end{pmatrix}, & G &= \begin{pmatrix} -m_b g \cos \psi \\ 0 \\ m_b g r \sin \psi \end{pmatrix}, \\ C &= \begin{pmatrix} 0 & 0 & -m_b r \dot{\psi} \\ 0 & b_m & 0 \\ 2m_b r \dot{\psi} & 0 & 0 \end{pmatrix}, & B &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \end{aligned} \quad (3.31)$$

3.3.2 Spin During the Flight

The spin of the ball does not appear in (3.30), since no contact couples it to the other coordinates and no torque acts about the centre of the ball. It is however still relevant, since the ball carries it through the whole flight and lands with this spin on the next surface. The spin is assumed constant during free flight [2]. Denoting by $\dot{\varphi}^-$ the value taken just before a transition and by $\dot{\varphi}^+$ the value just after it, as everywhere in this work, the spin the ball leaves with at liftoff is the one it arrives with at the next impact,

$$\dot{\varphi}(t) = \dot{\varphi}^+|_{t_{lo}} = \dot{\varphi}^-|_{t_{imp}}, \quad t \in [t_{lo}, t_{imp}], \quad (3.32)$$

where t_{lo} and t_{imp} are the instants of liftoff and impact. The spin must be carried alongside the state during this mode, and it enters the state again after the impact.

3.4 Hybrid System Formulation

The four modes derived above describe the motion when the ball is in one of them. The passages from one to another require two other elements: a condition marking the instant of the transition, called a guard, and a rule that gives the state immediately after reaching this condition, called a reset map.

Six transitions exist in this model, and each of them carries its own guard γ_i , following the notation of the reference work [2]. Every guard is built so that it stays negative while the mode is active and vanishes at the instant of the transition, which is what allows a solver to locate the crossing by a change of sign. Each transition is stated below, with the physical situation it describes, the guard that detects it and the reset map that follows. The whole is then assembled into a formal hybrid system.

3.4.1 Losses of Contact

A surface holds the ball only by pushing on it, so the ball leaves as soon as that push would have to become a pull. Consider a ball rolling at the distance R_{eff} from the axis, so that r is constant and $\dot{r} = 0$. Projecting the second law of Newton on the radial direction, the centripetal acceleration is produced by the radial component of gravity, $g \cos \psi$ directed outwards, and by the normal force N , directed inwards on a concave surface and outwards on a convex one,

$$N = \sigma m_b (g \cos \psi + R_{\text{eff}} \dot{\psi}^2), \quad (3.33)$$

with σ taken from Table 3.2. Contact requires $N \geq 0$, so the guard of every loss of contact is the negative of that force.

- **Outer hoop to free flight.** The surface is concave, and the ball leaves when

$$\gamma_1 = -(g \cos \psi + R_{\text{eff}} \dot{\psi}^2) = 0, \quad R_{\text{eff}} = R_o - R_b. \quad (3.34)$$

Near the bottom of the hoop, gravity and the centripetal requirement both press the ball against the track. Near the top, $\cos \psi$ becomes negative and the two compete, so the ball stays only if it moves fast enough.

- **Inner hoop, concave surface, to free flight.** The same expression applies, with the effective radius of that surface,

$$\gamma_2 = -(g \cos \psi + R_{\text{eff}} \dot{\psi}^2) = 0, \quad R_{\text{eff}} = R_{i,\text{in}} - R_b, \quad (3.35)$$

the smaller radius making the centripetal term grow faster for a given angular velocity. This surface carries a second guard, described below.

- **Inner hoop, convex surface, to free flight.** The surface is convex and pushes the ball outwards, which reverses the sign,

$$\gamma_3 = g \cos \psi + R_{\text{eff}} \dot{\psi}^2 = 0, \quad R_{\text{eff}} = R_{i,\text{out}} + R_b. \quad (3.36)$$

The situation is the reverse of the concave case: the ball is held at the top of the hoop, where gravity presses it against the wall, and leaves as soon as it acquires enough speed.

A fourth condition applies on the concave surface of the inner hoop alone. The wall is interrupted by the opening, and where the wall is absent there is no contact to lose. The ball then enters free flight whatever the value of (3.35), as soon as it reaches the angular span of the opening,

$$\gamma_{\text{hole}} = \frac{1}{2} \Delta \psi_{\text{hole}} - |\text{wrap}(\psi - \theta - \psi_{\text{hole}})| = 0, \quad (3.37)$$

where wrap maps an angle onto $[-\pi, \pi]$ and the difference $\psi - \theta$ is the angular position of the ball expressed in the frame of the hoops, in which the opening is fixed. This guard does not open a transition of its own. It decides which of the two surfaces of the inner hoop the ball can reach, since the concave one is accessible only through the opening, and it also removes the support from a ball already rolling on that surface.

No impulsive force acts at any of these transitions, so every state variable is continuous across them,

$$\boxed{r^+ = R_{\text{eff}}, \quad \dot{r}^+ = 0, \quad \theta^+ = \theta^-, \quad \dot{\theta}^+ = \dot{\theta}^-, \quad \psi^+ = \psi^-, \quad \dot{\psi}^+ = \dot{\psi}^-} \quad (3.38)$$

the radial velocity being zero because r was constant during rolling and the ball leaves tangentially. The spin also passes through unchanged, and its value at that instant, given by the rolling constraint (3.8) evaluated on the last contact state, is the one carried through the flight according to (3.32).

3.4.2 Impacts

A ball in free flight reaches a surface when its radial distance meets the corresponding effective radius. Three surfaces are available, and the ball arrives on each of them from a different side.

- **Free flight to the outer hoop.** The ball travels outwards and meets the concave surface from the inside,

$$\gamma_4 = r - (R_o - R_b) = 0. \quad (3.39)$$

This surface is continuous over its whole circumference, so no further condition applies.

- **Free flight to the inner hoop, concave surface.** The ball is inside the inner hoop and meets the wall from within,

$$\gamma_5 = r - (R_{i,\text{in}} - R_b) = 0, \quad (3.40)$$

subject to $\gamma_{\text{hole}} < 0$ at that instant, since the wall is absent over the span of the opening and the ball then continues its flight.

- **Free flight to the inner hoop, convex surface.** The ball travels inwards and meets the wall from outside, which reverses the sign,

$$\gamma_6 = (R_{i,\text{out}} + R_b) - r = 0, \quad (3.41)$$

subject to the same condition on the opening.

An impact differs from a loss of contact in that the velocity of the ball changes abruptly, and the equations of motion, which describe a flow, say nothing about an instantaneous event. The state after the shock follows from a reset map, which is a modelling choice.

The map used here treats the impact as perfectly inelastic in the normal direction and perfectly adherent in the tangential one. The radial velocity is annihilated, the ball being snapped to the surface, and the tangential motion is redistributed so that rolling is established immediately. The redistribution follows from the rotation the ball carries about its own centre. Writing that the spin after the impact must satisfy the rolling constraint (3.8), and that the ball arrives with the spin (3.32) it kept through the flight,

$$\boxed{\begin{aligned} \dot{\psi}^+ &= \dot{\psi}^- + \left(\dot{\theta}^- - \sigma \frac{R_b}{R} \dot{\varphi}^- \right), \\ r^+ &= R_{\text{eff}}, \quad \dot{r}^+ = 0, \quad \theta^+ = \theta^-, \quad \dot{\theta}^+ = \dot{\theta}^-, \quad \psi^+ = \psi^- \end{aligned}} \quad (3.42)$$

the added term measuring the mismatch between the rotation the ball carries and the rotation the contact requires, and the angular velocity of the hoops being unaffected, since the inertia of the hoop assembly is large compared with that of the ball.

3.4.3 Complete System

The four modes (3 in rolling, 1 in free-flight) and the transitions derived so far form a hybrid dynamical system, described by the quintuplet $\mathcal{H} = (Q, \mathcal{X}, f, \mathcal{G}, \mathcal{R})$ [14], [15]:

- the discrete modes $Q = \{q_1, q_2, q_3, q_4\}$, where q_1 is rolling on the outer hoop, q_2 is free flight, q_3 is rolling on the concave surface of the inner hoop and q_4 is rolling on its convex surface,
- the continuous state space $\mathcal{X} \subset \mathbb{R}^6$, whose elements are the vectors (3.1), together with the spin carried during q_2 ,
- the continuous dynamics f , given by (3.5) with the matrices (3.23) for the rolling modes and (3.31) for free flight,
- the guard conditions $\mathcal{G} = \{\gamma_1, \dots, \gamma_6, \gamma_{\text{hole}}\}$, given by (3.34) to (3.41) and (3.37),
- the reset maps $\mathcal{R}$, given by (3.38) for a loss of contact and by (3.42) for an impact.

The complete state of the system is the pair formed by the continuous state and the active mode. Between two transitions the system follows the dynamics of that mode, and when a guard vanishes the corresponding reset map is applied, the mode changes, and the evolution continues under the new dynamics. The resulting structure is shown in Figure 3.2.

The six guards reduce to two expressions once the sign σ of Table 3.2 is used to carry the curvature of the surface. Every loss of contact is written as $\gamma = -\sigma (g \cos \psi + R_{\text{eff}} \dot{\psi}^2)$, which is the normal force (3.33) taken with the opposite sign, and every impact as $\gamma = \sigma (r - R_{\text{eff}})$, the sign expressing that the ball reaches a concave surface from the inside and a convex one from the outside.

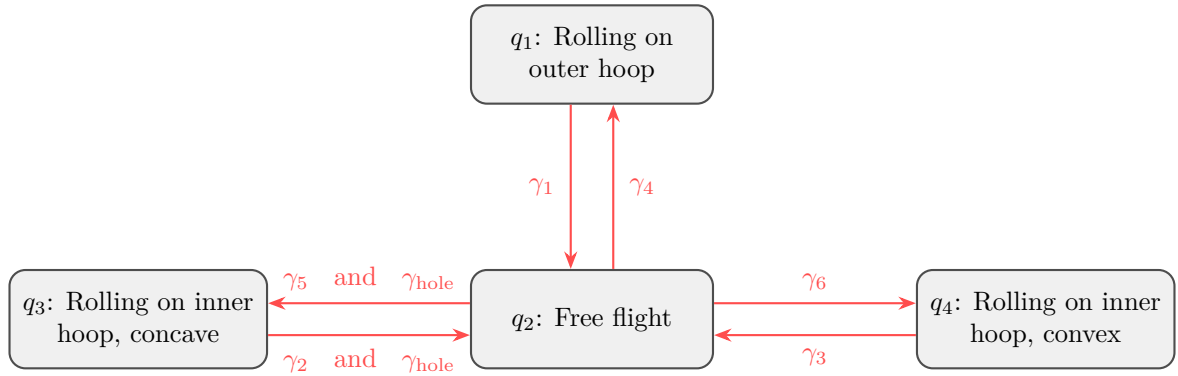


Figure 3.2: Hybrid automaton of the ball-in-double-hoop system. Each arrow is triggered when the corresponding guard vanishes. Losses of contact are governed by γ_1 , γ_2 and γ_3 , impacts by γ_4 , γ_5 and γ_6 , and the passage over the opening of the inner hoop by γ_{hole} .

Free flight is the only mode connected to all the others, which reflects the physical arrangement of the apparatus. The ball can reach a surface only by travelling through the air, and can leave one only in the same way, so that the three rolling modes are never adjacent and no transition connects them directly.

3.5 Numerical Implementation

The hybrid model derived above is implemented in MATLAB. This implementation is used throughout the rest of this work, to size the components, analyse the identified parameters, design the test controllers and compare them with the results obtained on the real plant. Its purpose is therefore not only to reproduce the equations, but to serve as a reference against which the prototype is compared.

Two requirements follow from this purpose: the same code must simulate the system in open loop and in closed loop, and each run must be saved, to remain comparable with a measurement. This section reports the choices made to satisfy these requirements, the open-loop results they produce, and the limitations of this model.

3.5.1 Implementation

The equations of motion of the active mode are integrated with the `ode15s` solver of MATLAB [26]. Two families of solvers exist for this task.

- An explicit method computes the next state directly from the current one, which is cheap per step but forces the step to shrink whenever a fast dynamic is present, whatever its amplitude, since a step too large for that dynamic makes the integration diverge.
- An implicit method instead solves an equation at each step to find a state consistent with the dynamics at the end of the interval, which costs more per step but remains stable with large steps.

The second family is the one suited to stiff problems, that is to problems containing dynamics evolving on very different time scales, where an explicit method would spend its effort resolving a fast transient that has long since died out. The rolling dynamics alone are not stiff. What makes the implicit solver the right choice is the closed-loop use the implementation is written for: the controllers add the loop of the driver, whose dynamics are an order of magnitude faster than the ball, and the transitions can bring abrupt changes of regime. The solver is therefore chosen once for the hardest case rather than switched by scenario, the cost on the passive runs being a fraction of a second either way. The solver returns a continuous representation of the solution, so a trajectory can be evaluated at any instant, and compared to a measured trajectory point by point.

Transitions are located by the event mechanism of the solver, with one function declared per guard. When an event is triggered, the integration stops and the reset map is applied to the last recorded state. The integration then resumes in the new mode.

The hole is taken into account by adding the hole condition in the event detection, so that no guard fires while the ball faces the opening. At the instant of a crossing, the state sits exactly on the guard. In practice, the integration resumes slightly past it, so that the same event is not detected infinitely. The spin of the ball is stored at every loss of contact, and recovered at the next impact.

Each run is finally stored with its trajectory, the mode intervals and the parameters that produced it, so that the different figures can be traced back.

3.5.2 Open-Loop Results

The implementation is used on a set of open-loop scenarios, each declaring an initial mode and state, with $\tau = 0$. Two verifications are made on them: a behaviour verification to check that the expected modes, transitions and states are active, and an energetic one, that verifies if the total energy stays constant if no dissipating mechanism is left in the process (with $b_b = b_m = 0$ and no torque applied).

Six scenarios are reported in [Figure 3.3](#). Two exercise a rolling mode on its own, and the four others cross at least one transition. All transitions and modes are used in the present scenarios.

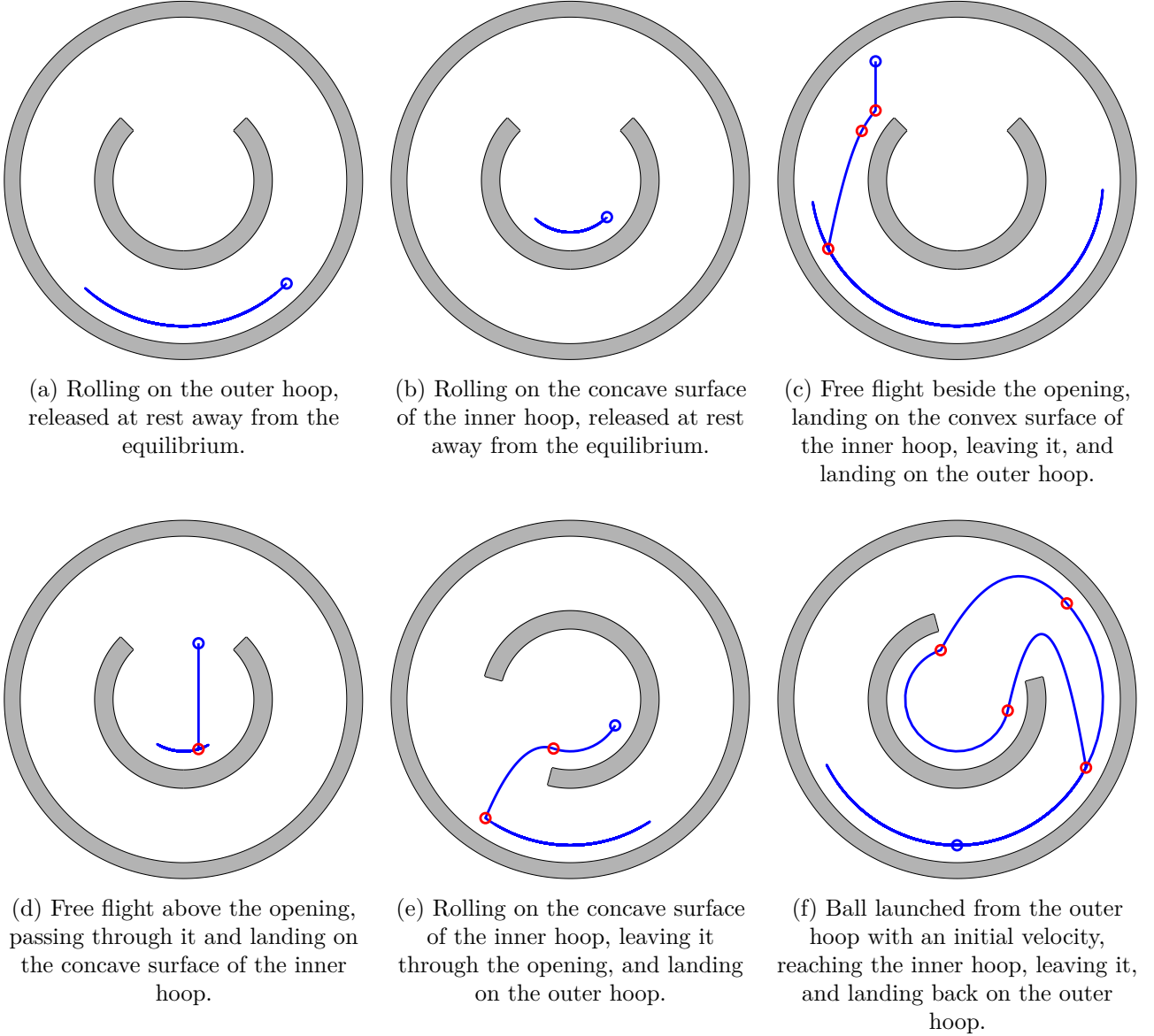


Figure 3.3: Open-loop trajectories of the six scenarios, released with $\tau = 0$. The blue dot marks the initial position of the ball and the blue line its trajectory. Red dots mark the positions at which a transition is triggered. The hoops are drawn at their starting orientation.

Each mode is reached, and every transition is triggered where it should be. The six scenarios show every transition of the model at least once. The guard bounding the opening hole is seen twice: when it lets the ball in, and when it lets it out. The losses of contact preserve the state, and the landings are located within 6×10^{-14} m of the geometric contact radius, so that the reset map applies to a state that already satisfies the guard, rather than correcting it.

The energetic tests meet the expectations: the relative deviation stays below 0.013 % over three seconds and below 0.026 % over ten. The energy changes by less than 1×10^{-7} J across a loss of contact, and it always decreases across an impact.

This decrease at impacts is large, between 91 % and 99 % carried into the impact. This follows directly from the two assumptions of the reset map. The radial velocity, mostly corresponding to the velocity at the end of a fall, is entirely absorbed. What remains of the tangential motion is also reduced by the establishment of rolling, leading to this decrease. This means that the simulation is optimistic about captures, and cannot directly discriminate between a ball caught and a ball that would have bounced away. If the prototype shows a ball bouncing or sliding on landing, introducing a restitution coefficient is one of the first changes to make [27], [28].

Dissipation itself can also be a limiting factor of this simulation: it is carried by two viscous coefficients alone, while other mechanisms are considered negligible or not represented in a differentiable model (see [Appendix A](#)). Their effect can appear, and appear in the error to be compensated.

The ball is finally treated here as touching a cylindrical surface at a single point. In practice, some tracks hold the ball on two rings. The resulting geometry is properly derived once the rails are selected (see [section 5](#)).

Apart from these limitations, the model is now complete and its implementation validated. It is written in terms of quantities taken from a reference design rather than from an apparatus that exists [2]. The chapters that follow show the components selection and parameters identification that allow a proper comparison with the simulation.

4 System Design

The two previous chapters described what a ball-and-hoop apparatus is and derived the equations governing it. Neither of them concerns a physical object: the state of the art reports what others have built, and the model is written in terms of masses, radii and friction coefficients that no prototype has yet supplied. This chapter focuses on selecting components that give those symbols a proper value.

The selection follows two series of requirements. First, the apparatus must be able to perform the four tasks this work sets out to demonstrate, which bounds the torque, the resolution and the timing of the whole chain. Second, the apparatus must keep valid the assumptions on which the mathematical model rests, or ensure that corrections are applicable, since a component that violates one of them makes the model useless. This section derives the different requirements, before describing the components selection and assembly.

4.1 Requirements

Four tasks of increasing difficulty are set as objectives for the prototype. They are formalised as control problems in [section 6](#), and each of them demands a performance that directly influences the selection of the components.

- T1 Regulation.** The ball is held at one of the equilibria of the system, against small perturbations. The demand is on measurement rather than on power, since a static equilibrium requires no sustained effort, and the resolution on the ball position sets the smallest deviation the controller can react to.
- T2 Tracking.** The ball follows a time-varying reference within the same rolling mode. The reference now moves, so the loop must be fast compared with the motion it is asked to reproduce, which sets the sampling frequency of the whole chain and bounds the acceptable latency of the sensors.
- T3 Loop-the-loop.** The ball completes a full revolution while keeping contact with the track. Contact at the top requires the ball to carry enough speed for the surface to keep pushing on it, and that speed must be built up against gravity, which sets the peak torque the actuator must deliver. The contact must also sustain the tangential force that rolling demands, which bears on the surface of the tracks rather than on the motor.
- T4 Flying ball.** The ball leaves one surface, travels through the air and is caught on another. During the flight the ball is mechanically disconnected from the hoops, so any sensor relying on that contact reports nothing at the moment the information matters most. This task alone rules out every contact-based measurement of the ball position.

A second set of requirements comes from the model itself. Several of the assumptions on which [section 3](#) rests are properties of the hardware rather than of the mathematics. Each of them is therefore raised where the component that protects it is selected.

These demands are not independent. This is why the selection is made iteratively, each component being chosen against the requirements and against the components already selected.

4.2 Architecture

The apparatus is organised into four subsystems, shown in [Figure 4.1](#). The controller evaluates the control law and issues a command, the actuation system converts it into a torque applied to the hoops, the mechanical assembly carries the ball, and the sensing system returns the measurements the controller needs.

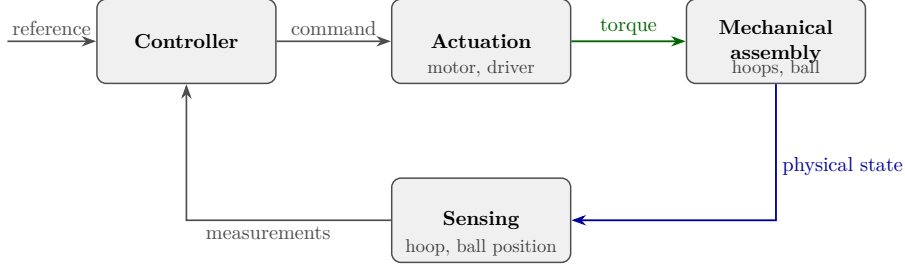


Figure 4.1: Block diagram of the apparatus, showing the four subsystems and the signals exchanged between them.

The loop closes on the ball position, which is the quantity to be controlled and the one that fixes the requirements on everything else. A precision of $\Delta\psi_{req} = 1^\circ$ is retained as the target, and the resolution of both sensors is derived from it.

The rest of this chapter follows the subsystems in the order in which they constrain one another. The mechanical components come first, since they fix the geometry from which every other requirement is computed. The drive system follows, sized against the most demanding task. The sensors and the embedded computer come last, their requirements being expressed in terms of the geometry and of the dynamics that the previous choices have fixed.

4.3 Component Selection

Each component is selected against the requirements stated above, and against the components already retained. For each of them, the constraints are expressed, quantified where it is possible, and confronted with a small number of candidates.

4.3.1 Mechanical Components

Ball

Three properties are required of the ball, and candidates are compared in [Table 4.1](#):

- its mass must be uniform, since the model uses the inertia of a homogeneous sphere
- its contact with the track must be dominated by rolling rather than by deformation, since the model carries a viscous coefficient and no viscoelastic term,
- its diameter must be small enough to pass through the opening of the inner hoop.

	Table tennis	Rubber	Chrome steel
Diameter	40 mm [29]	25 mm	25 mm
Mass	2.7 g [29]	≈ 11 g	67 g
Mass distribution	Hollow shell	Uniform	Uniform
Young modulus	Low	10 MPa to 50 MPa [30]	≈ 200 GPa

Table 4.1: Comparison of the ball candidates.

The solid chrome steel ball is retained. It is rigid enough for its deformation at the contact to be neglected, its mass is uniformly distributed, and its diameter passes through the opening. The table tennis ball fails on two counts, being a hollow shell whose mass sits near its surface, and being light enough to bounce and to be affected by aerodynamic effects. The rubber ball meets the geometric requirement but not the second one, its rolling behaviour being dominated by viscoelastic losses that no single coefficient describes.

Tracks

Two assumptions of the model bear on the surface the ball rolls on, the no-slip assumption and the planar motion assumption. The first holds only if the contact can supply the force the motion demands, and the second only if the ball is constrained laterally. A bare printed surface answers neither well, which is why dedicated tracks are added to the design.

The friction coefficient needed to sustain rolling follows from the reduced model. It is the ratio between the tangential force demanded by the spin of the ball and the normal force actually available,

$$\mu_{req}(t) = \frac{F_t(t)}{N(t)} = \frac{\frac{2}{5}m_b |R_o\ddot{\theta}(t) - R_{eff}\ddot{\psi}(t)|}{m_b (g \cos \psi(t) + R_{eff}\dot{\psi}(t)^2)}. \quad (4.1)$$

This ratio varies along a trajectory and grows sharply wherever the normal force becomes small, which is what happens near the top of the hoop. Dry PLA on steel offers $\mu \approx 0.52$ to 0.67 [31], against $\mu \approx 0.6$ to 0.86 for rubber on steel [30].

Rubber O-rings of an NBR70 compound are therefore retained as tracks, two per rolling surface. They raise the available friction and hold the ball in a vee that keeps it in the plane of the hoops. Their cord section and their diameters are taken from commercially available standard sizes, so that a damaged ring can be replaced without reprinting anything.

Carrying the ball on two rings rather than on a cylindrical surface has a consequence for the model. The contact occurs at two points instead of one, which displaces the centre of the ball and changes the radius at which it rolls. The correction is derived before any experiment is performed on the plant (see section 5).

Hoops

The hoops fix the geometry of the whole apparatus, and they are dimensioned around the two choices above. They must receive the rings retained as tracks, leave the ball enough room to pass through the opening of the inner hoop, and fit the build volume of the printers available for this work. They must also present a rolling surface whose radius is known accurately, since any error on it propagates into every equation of the model.

Both hoops are printed in PLA by fused deposition modelling, from a parametric model. Printing rather than machining is retained because the geometry is not a simple revolution: the grooves receiving the rings, the opening and the mounting pattern are produced in one operation, and a change of dimension costs a reprint rather than a new fixture. The dimensions of the reference project are taken as a starting point, and adapted to the ball and to the standard ring sizes retained above. The resulting design is shown in Figure 4.2.

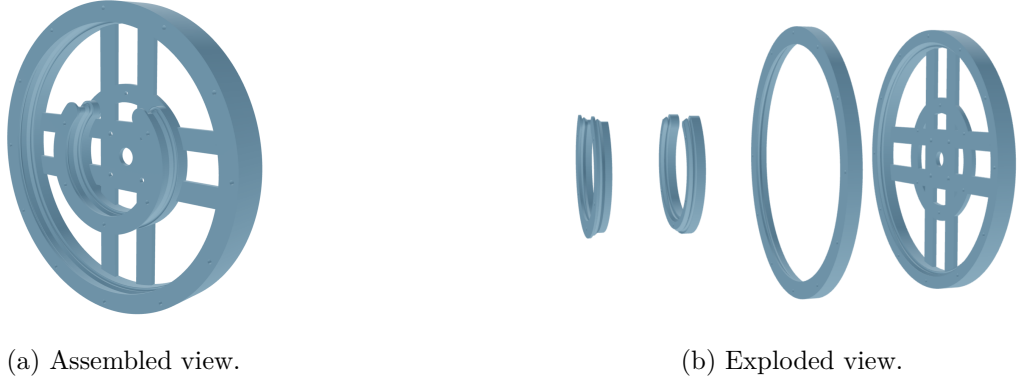


Figure 4.2: CAD model of the two hoops, showing the tracks and the opening of the inner hoop.

4.3.2 Drive System

Motor

Two families of actuators are commonly used for this kind of application. Stepper motors offer a simple open-loop control, but their torque drops quickly with increasing speed and they are subject to resonance and to step loss under dynamic loads [32]. Brushless DC motors offer a better power-to-weight ratio and a higher efficiency, and produce a continuously variable torque across their operating range when driven in closed loop by Field-Oriented Control, described in [Appendix B](#). The tasks of this work require a smooth torque rather than discrete positioning steps, so a BLDC motor is retained.

The torque the motor must deliver is set by the most demanding task, the looping. To perform it, the gravitational load on the ball must be compensated. The rolling constraint reflects this load onto the hoop. Evaluated at the worst-case ball position, $\sin \psi = 1$, it gives

$$\ddot{\theta}_{req} = \frac{5}{2} \cdot \frac{g R_{eff}}{R_o (R_o + R_b)} \approx 144 \text{ rad s}^{-2}, \quad \tau_{req} = I_h \ddot{\theta}_{req} \approx 0.525 \text{ N m}. \quad (4.2)$$

On an actual trajectory, this maximal contribution is reached only for an instant, which makes this estimate an upper bound on the peak torque rather than a value the motor must continuously sustain. The trajectory optimisation (see [section 6](#)) returns a maximal peak of 0.473 N m for the same manoeuvre, which confirms that the bound is conservative and that it is the right order of magnitude. Both values rest on the idealised contact geometry. The use of tracks lowers the effective radius and raises the rotational inertia that this contact must accelerate. The corrected geometry is derived once the rings are measured (see [section 5](#)): the value moves slightly, but leaves the selection unchanged.

On this basis, the ODrive D5065 270KV motor is retained. Its rated torque of 0.612 N m at 20 A covers the bound comfortably [33], and it is the actuator used in the prototype distributed with the reference project [21]. Its model and characteristics are detailed in [Appendix C](#).

Driver

The driver must implement the Field-Oriented Control algorithm, read the angular position of the rotor, accept commands from the outer controller, and convert them into phase currents. The ODrive S1 is recommended by the manufacturer for the D5065 and implements all of these functions, so its selection is straightforward [34].

The driver also determines whether the input assumed by the model is realisable. The model com-

mands the angular acceleration of the hoops rather than a torque, following the idea that the driver reaches the commanded quantity faster than the mechanical response of the system. This assumption is a property of the velocity loop of the driver, which must be verified rather than assumed. The inner loop is closed internally in the driver, at a rate set by its own tuning. The resulting cascade architecture is the one adopted throughout this work, and the measurement establishing the separation of time scales is reported with the control design (see [section 6](#)).

Power supply

The ODrive S1 accepts a DC supply voltage between 12 V and 50.5 V [\[34\]](#). A 24 V supply is used, comfortably inside that range and below the 32 V rating of the D5065 itself. The phase current required during the most demanding manoeuvre is estimated from the numerical simulation at 14 A to 16 A, so two supplies are connected in parallel, providing up to 20 A combined.

4.3.3 Sensors

Two quantities must be measured, the hoop angle θ and the ball angular position ψ . Both sensors are selected against the precision target set earlier, $\Delta\psi_{req} = 1^\circ$ on the ball position, which is the quantity the controller acts on and the one from which every other requirement follows.

Hoop angle

The hoop angle is measured by an encoder reading the rotation of the motor shaft, the hoops being rigidly attached to it. The required resolution follows from the target on the ball position. At the contact point, an angular step $\Delta\theta$ of the hoop produces a linear displacement $R_o \Delta\theta$, while the same displacement seen by the ball corresponds to an angular step evaluated at the effective radius of the outer track, $R_{eff} = R_o - R_b$. Requiring the contribution of the hoop to match the target gives

$$\Delta\theta_{req} = \frac{R_{eff}}{R_o} \Delta\psi_{req} \approx \frac{100}{112.5} \times 1^\circ \approx 0.89^\circ. \quad (4.3)$$

The AMT102-V encoder, recommended by the manufacturer alongside the motor and driver already retained, provides 2048 pulses per revolution in quadrature, that is 8192 counts per revolution [\[33\]](#). Its resolution is therefore $360/8192 \approx 0.044^\circ$, roughly twenty times finer than required. Such a margin is welcome, because this measurement is also used to estimate the hoop velocity by differentiation, and a coarse resolution would make that estimate noisy.

Ball position

The prototypes reviewed earlier use various sensors for this measurement, from infrared arrays to encoders coupled to the ball itself (see [section 2](#)). All of them rely on the ball touching a surface, and therefore report nothing during the free-flight phase. A contactless sensor is the only option compatible with the four tasks, and a Raspberry Pi Camera Module V2 is therefore retained.

This choice has a consequence on the rest of the design: acquiring and processing an image is far more expensive, in memory and in computation, than reading an encoder. This can introduce latency into the control loop, and the image can only be treated by a computer powerful enough to handle image processing. The latency is an important measure of the assembled prototype (see [section 5](#)).

The vision pipeline detects the ball by colour segmentation and runs entirely on the embedded computer. It is adapted from the code distributed with the reference project rather than derived from

scratch [21], and rewritten for the current versions of the libraries it depends on. Each frame is cropped, converted to a colour space robust to lighting variations, and thresholded to isolate the pixels corresponding to the ball. The centre of the resulting mask gives the position of the ball in pixels, and its offset from the optical centre gives its coordinates (u, v) in the pixels plane, which can then be translated into metric values $(u_{[m]}, v_{[m]})$ and converted into the angle ψ by the relation $\psi = \text{atan2}(v_{[m]}, u_{[m]})$. The complete procedure is reported in appendix (see [Appendix D](#)).

Two properties of this measurement matter for the design. If the optical axis of the camera remains perpendicular to the hoop plane and passes through the rotation axis, the measurement is independent of the focal length and the working distance. Its precision depends on the number of pixels occupied by the ball in the image, and therefore on the distance at which the camera is placed. Both conditions are enforced during the assembly.

4.3.4 Embedded Computer

The last component is the computer. At every control step, it must acquire a frame and extract the ball position from it, read the hoop angle, estimate the two velocities needed by the controller, evaluate the control law, and transmit the resulting command to the driver. All of this must fit within one control period, which is the quantity to be fixed first.

The period is set by the dynamics the loop must follow: the motion of the ball. Linearising the rolling dynamics around the stable equilibrium leads to the natural frequency of the ball oscillating in the hoop. The inner hoop ($R_{\text{eff}} = 32$ mm) being smaller, its natural frequency is higher, and is the most restricting one for the design choices:

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{g}{R_{\text{eff}}}} \approx 2.8 \text{ Hz}, \quad (4.4)$$

with the idealised geometry retained so far. Adapting it to the exact geometry with tracks does not change the frequency too much.

A common rule places the sampling frequency between ten and twenty times the dominant dynamics [35], leading to 28 Hz to 56 Hz. A control frequency of 50 Hz is adopted in this work, as in the reference project [2], leaving 20 ms for the whole chain.

The most restricting requirement on the computer itself is linked with the vision pipeline, once again. It needs several hundred megabytes of working memory and must be compatible with libraries such as OpenCV. A Raspberry Pi 3B+ is therefore retained, with its 1 GB of memory. It runs Python natively, and supports the camera through a dedicated interface [36]. Whether the different operations actually fit within the 20 ms budget depends on the resolution and on the implementation. It is measured on the assembled prototype (see [section 5](#)).

4.4 Assembly

Selecting the components fixes the apparatus on paper, but building it raises a separate question, since several of the assumptions the model rests on are not properties of the components themselves, but of the way they are assembled. The planar motion assumption, in particular, depends on a well-chosen ball and a well-printed hoop but, more importantly, can only hold if the rotating assembly turns about

a fixed axis. The assembly is therefore a crucial step in protecting the model.

4.4.1 Mechanical Assembly

Both hoops are produced by fused deposition modelling in PLA. Each is printed in two halves, split by a horizontal plane passing between the two grooves of a track. Printing a downward-facing groove would require support material inside the rolling surface itself, and would leave a surface finish incompatible with the rolling assumption. The halves are assembled with M3 screws.

The ball is carried by two O-rings per track, axially separated by 14.7 mm. Three pairs are fitted in total, one for the outer hoop and two for the inner one, one on each of its faces. Since the inner hoop is an open profile of 270° , its rings cannot be fitted whole. They are cut and joined with cyanoacrylate, the junction being placed on a portion of low curvature so that the weakened section carries as little stress as possible.

The hoops are not mounted directly on the motor shaft. A printed adapter is used instead, consisting of a collar engaging the front shaft and a flange bolted to the M3 pattern already carried by the hoops, clamped by an M3 screw threaded into a brass insert.

The prototype is built on an aluminium plate carrying an M6 hole grid of 25 mm pitch. The stator is clamped by a printed support bolted to that grid through L-shaped brackets, setting the rotation axis at 133.5 mm above the plate. The encoder is mounted on the rear shaft, the motor being a double-shaft machine, so that the measurement is taken on the same rigid body as the hoops. Every element of this chain bears on the planar motion assumption, since a hoop mounted eccentrically, a collar loose on the shaft or a support that flexes under the reaction torque all produce an out-of-plane component that the model does not carry. The three printed supports are shown in [Figure 4.3](#).

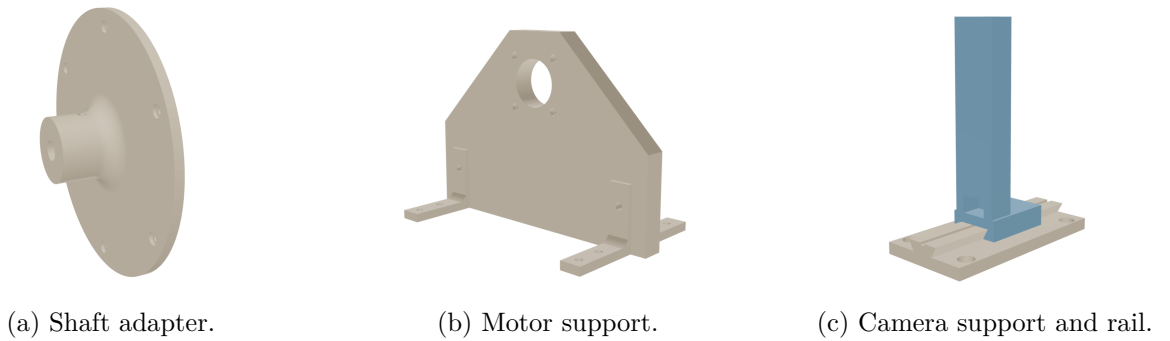


Figure 4.3: Printed parts holding the apparatus.

4.4.2 Camera Support

The measurement of ψ is independent of the focal length, but only if the optical axis is perpendicular to the hoop plane and passes through the rotation axis, as established when the sensor was selected. The support enforces both conditions by construction. Every degree of freedom left to it is a potential source of misalignment, so all are removed except the working distance d_{cam} , which sets the fraction of the image the apparatus occupies and therefore the number of pixels available to locate the ball. The optical axis is fixed at the height of the rotation axis, and the support slides on a rail bolted to the grid through a 60° dovetail. The travel of that rail follows from the field of view of the sensor.

The vertical field of view is the more restrictive, at 48.8° , and an object of diameter D fills a fraction κ of the image at the distance

$$d_{cam} = \frac{D}{2 \kappa \tan(\text{FOV}_v/2)}. \quad (4.5)$$

With an outer diameter of 237 mm, full coverage requires 261 mm. This extreme is not desirable, leaving no margin around the hoop which makes the rim difficult to detect during the sensor commissioning. The rail provides 77 mm of travel, corresponding to $d_{cam} \in [266, 344]$ mm, which spans the useful range of coverage.

4.4.3 Electrical Assembly

The electrical assembly connects three chains, the actuation chain comprising the motor, the driver and the encoder, the vision chain comprising the camera and the computer, and the supplies feeding each of them. The architecture is shown in Figure 4.4.

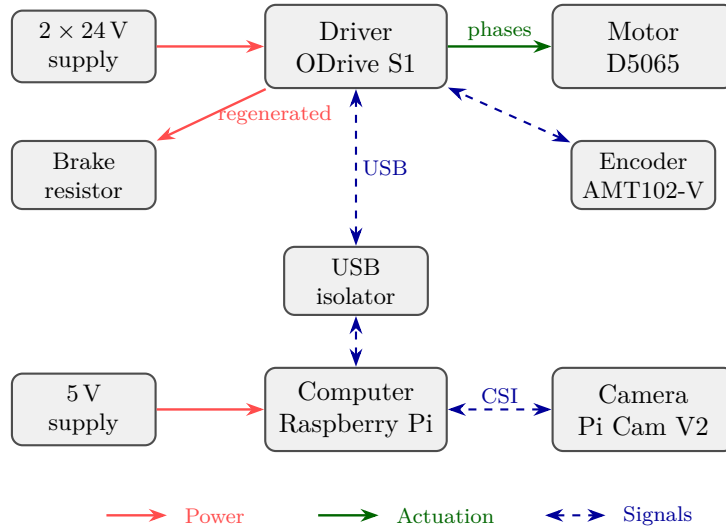


Figure 4.4: Electrical architecture of the prototype.

Three connections require care:

- The driver receives a brake resistor of $2\ \Omega$ and 50 W, which is not optional here: whenever the hoop decelerates, the motor regenerates energy onto the DC bus, and without a path to dissipate it the bus voltage rises until an over-voltage fault trips the driver [34]. Reversing torques occur during every task of this work, so the condition arises regularly rather than exceptionally.
- The encoder cable is supplied for an earlier ODrive board, so its pinout is adapted manually against the datasheet of the S1 [34].
- The computer communicates with the driver over USB, and the documentation states that USB and the DC input cannot be used simultaneously without an isolator, the two supplies otherwise sharing a ground path through the link [34]. An isolator is therefore fitted.

5 Parameter Identification

The previous chapter describes how the apparatus is built. Using it requires one more step: the model is written in terms of physical parameters that the design alone does not supply. Some of them are known by construction, such as the radii of the hoops or the inertia of the ball, but others depend on the specific components and assembly tolerances, and must be measured. They include friction coefficients and the inertia of the rotating assembly.

The identification is made in a fixed order, each step depending on the results of the previous one. The effective geometry is derived first, now that the tracks are chosen. The sensors are then commissioned, since nothing can be identified before the instruments themselves are trusted. Experiments follow, identifying the inertia and friction of the rotating assembly (E1) and the rolling friction of the ball (E2), and a last one validates the complete model (E3).

5.1 Effective Contact Geometry

The model treats the ball as rolling on a cylindrical surface at a single contact point, at a distance $R_{eff} = R_o - R_b$ from the axis, with rolling radius R_b . Neither holds exactly for the assembled prototype: the ball is carried by two O-rings per track, axially separated by h , and therefore touches the track at two points placed symmetrically about its mid-plane.

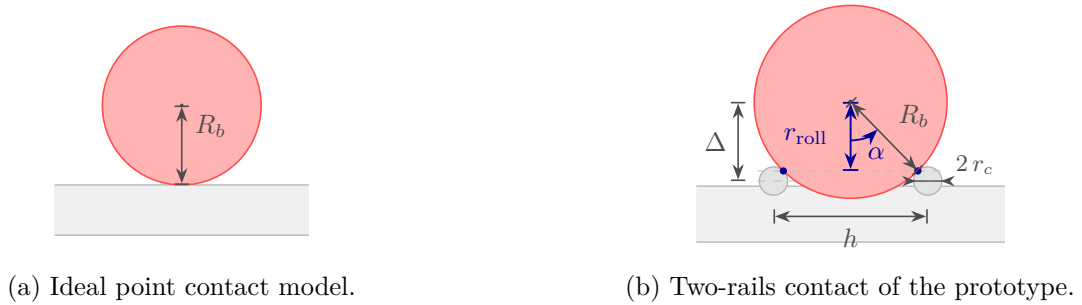


Figure 5.1: Contact geometry of the idealised model and of the real prototype.

With R_g the O-ring centreline radius and r_c the O-ring cord radius,

$$\Delta = \sqrt{(R_b + r_c)^2 - (h/2)^2}, \quad R_{eff} = R_g - \sigma\Delta, \quad (5.1)$$

with σ carrying the curvature of the surface, and giving $R_{eff} = 91.6$, 67.4 and 32.6 mm for the outer, inner outside and inner inside tracks respectively. The rolling radius, that is the perpendicular distance from a contact point to the spin axis of the ball, follows from the axial offset $a = R_b (h/2)/(R_b + r_c)$ of the contact points,

$$r_{roll} = \sqrt{R_b^2 - a^2} = R_b \frac{\Delta}{R_b + r_c} = 10.64 \text{ mm} < R_b, \quad (5.2)$$

corresponding to a contact half-angle $\alpha = \arcsin(a/R_b) = 31.7^\circ$.

The idealised model is recovered when $h \rightarrow 0$, since $r_{roll} \rightarrow R_b$. Physically, the two contact points define a rolling axis corresponding to a chord of the ball rather than a diameter, so the ball must spin faster than the ideal model predicts to travel the same distance. Its effective rotational inertia is therefore larger, by a factor

$$k = 1 + \frac{2}{5} \left(\frac{R_b}{r_{roll}} \right)^2 = 1.552, \quad (5.3)$$

against $1 + 2/5 = 1.4$ for the ideal case. All values are modified accordingly and used in every experiment in control design task (see [section 6](#)).

5.2 Sensor Commissioning

Every measurement reported below relies on the encoder for θ and on the camera for ψ . Neither can be trusted before it has been characterised, and the camera in particular requires more than reading its datasheet, since several of its properties depend on the assembled apparatus rather than on the sensor itself.

Encoder

The encoder is tested by driving the shaft through five complete turns against a hand-aligned reference mark, and comparing the reported count with the 40 960 counts expected. The relative error is 0.16 %, that is about 2.9° over 1800° .

This is sixty times the nominal resolution of 0.044° , and it comes from the alignment rather than from the sensor, since a mark positioned by eye cannot be trusted to better than a few degrees. The test therefore says nothing about the fineness of the encoder, which is a property of the device. What it does establish is that the count scales correctly: a wrong resolution setting or a single-edge rather than quadrature decoding would produce an error of a factor of two, not of a fraction of a percent. It also confirms that no pulse is missed over the travel, which is the only failure mode an incremental encoder exhibits in normal operation.

Acquisition parameters

The camera is placed at $d_{cam} = 330$ mm, within the range the rail allows. The acquisition settings retained are a resolution of 820×616 , an exposure of $18\,000\,\mu\text{s}$ and an analogue gain of 12.0, with the automatic white balance converged once and then locked, so that the colour of the ball does not drift between trials. At this distance the outer hoop spans about 546 px, which gives a vertical coverage of 88.5 % and a scale factor of $\lambda = 4.32 \times 10^{-4}$ m/px.

The segmentation follows the architecture of the reference implementation [21]: a cropping of the sides of the image, a scalar projection of the colour channels, a coarse search over the full frame, a fine tracking window, and the image moments of the retained mask. Three parameters of this sequence are set experimentally on the present apparatus. The ball is painted red, which separates it from the printed parts and from the aluminium plate in the projection. The threshold isolating it is set to 110, low enough to keep the ball fully detected under the lighting of the laboratory and high enough to reject the reflections on the rings. The tracking window is set to 64 px, a value discussed below.

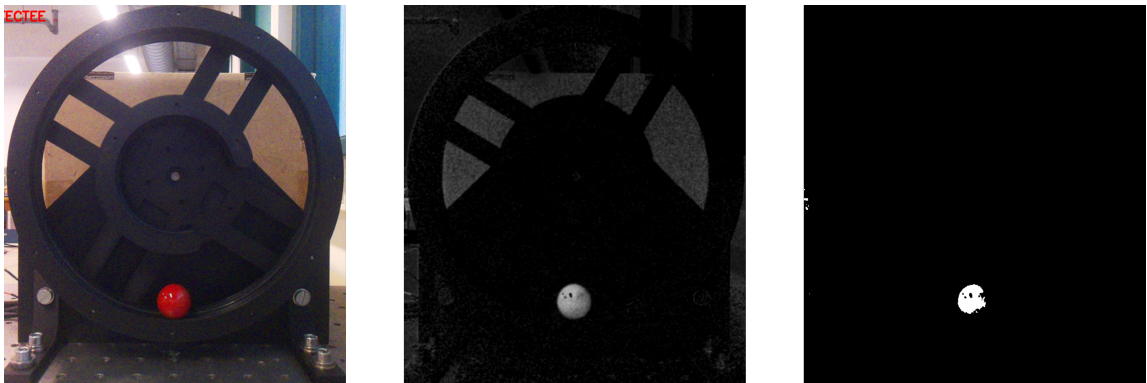


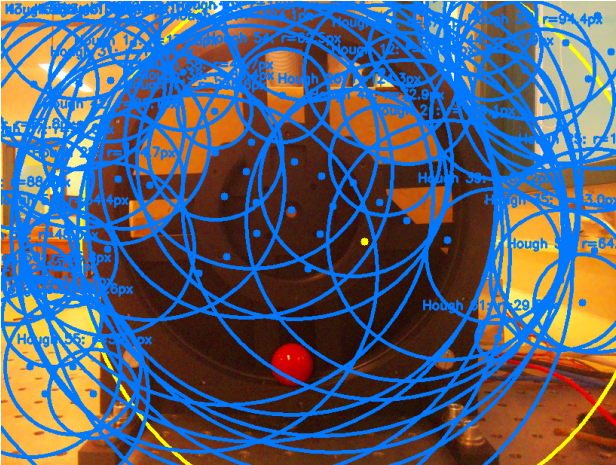
Figure 5.2: Single frame from the vision chain, with the raw image, the separation through the threshold and the detected ball centroid.

Reference centre

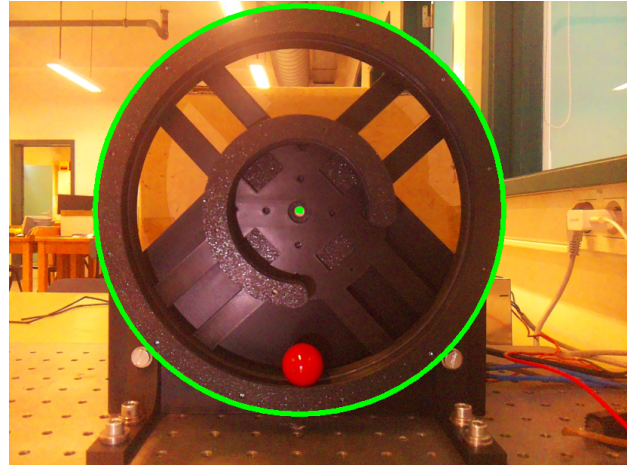
The angle ψ is computed from the offset between the ball centroid and a reference centre (c_x, c_y) , the projection of the rotation axis in the image. An error on this centre produces a bias on ψ that varies with the position of the ball, and no averaging over trials can remove it, since the same error is present in every frame.

That centre does not coincide with the centre of the image. The support constrains the camera to stay parallel to the plane of the hoops, which is the condition the measurement requires, but it does not centre the apparatus in the frame to the pixel. Such an offset displaces the projection of the rotation axis without distorting the angles, and is absorbed entirely by (c_x, c_y) , provided that centre is measured rather than assumed.

The rim of the outer hoop is detected by a Hough transform [37]. Applied without constraint, the transform returns dozens of circles per frame, responding as readily to the spokes, to the rings and to the background as to the rim itself, as Figure 5.3 shows. The rim is among them, but nothing selects it. Supplying the radius expected at the working distance, with a tolerance around it, removes the ambiguity: candidates far from that value are rejected before the centre is computed, and a single circle is retained.



(a) Without a radius constraint.



(b) With the expected radius imposed.

Figure 5.3: Circles returned by the Hough transform on the same view.

The constrained detection is repeatable. Over four consecutive runs the recovered centre varies by 2 px in u and 3 px in v , a root mean square dispersion of 1.6 px, and the detected radius by 1 px. Three of the four are identical to the pixel, the rim being a circle and its detection therefore independent of the orientation of the hoops. Propagated through the arc tangent at the orbit radius of the outer track, this dispersion gives an uncertainty of about 0.33° on ψ , within the target of 1° set during the design.

A second method was implemented while the first was still unreliable, and it is retained as an independent check. Rather than fitting the hoop, it fits a circle to a recorded trajectory of the ball, which describes an arc centred on the same axis, using the algebraic method of Taubin [38]. The two rest on different objects, one on the geometry of the apparatus and the other on the motion of the ball, so their agreement verifies the centre rather than repeating its measurement. The centres they return differ by 5.6 px, which is the same order as the dispersion of either fit.

Latency

The delay between the instant the ball occupies a position and the instant the controller receives it is not reported by any datasheet, since it accumulates over acquisition, transfer and processing. It is measured on the assembled chain, by commanding a step on the hoop at an instant known to the driver, recording when the encoder reports the motion, and comparing it with the timestamp of the frame in which the camera first sees the ball respond. Instrumenting the acquisition script also gives the duration of each stage taken separately, which shows where the delay accumulates. The total amounts to 19.8ms on average, that is roughly one control period.

The size of the tracking window matters here. Too large, and processing a frame takes longer than the control period, so the next acquisition is missed and the sampling becomes irregular. Too small, and the ball leaves the window whenever it moves fast, its displacement between two frames exceeding the half-width. A window of 64px is retained as a first choice, which keeps the period uniform over the identification campaign, and can be widened for the faster tasks of the control campaign if needed (see [section 6](#)).

5.3 E1 — Inertia and friction of the rotating assembly

The first experiment identifies the inertia I_h reflected at the motor axis and the friction acting on the rotating assembly. The ball is absent, so the hoops, the adapter and the rotor form a single body turning about a fixed axis, and the dynamics reduce to an inertia-friction model,

$$I_h \ddot{\theta} + b_m \dot{\theta} + \tau_c \text{sign}(\dot{\theta}) = \tau, \quad (5.4)$$

where τ_c is a Coulomb term, constant in magnitude and opposed to the motion. It is measured here even though the model does not carry it (see [section 3](#)), since knowing its magnitude is what allows its effect to be recognised in the results that follow. The torque is never measured directly. It is obtained from the current reported by the driver and the torque constant of the motor.

The identification is split in two. A series of steady-state measurements gives b_m and τ_c , and a free deceleration then gives I_h , the two other parameters being known by that point.

Friction from steady state

A series of constant velocities is imposed in velocity mode. Once the steady state is reached, the acceleration vanishes and [\(5.4\)](#) reduces to an affine relation between torque and velocity,

$$\tau = b_m \dot{\theta} + \tau_c \text{sign}(\dot{\theta}). \quad (5.5)$$

The two directions of rotation are fitted separately, since the Coulomb term changes sign with the motion. The slope of each regression gives b_m and the intercept gives τ_c , which yields two independent estimates of each parameter rather than one.

Samples below 60 rad s^{-1} are excluded. In that range the friction no longer follows the affine law of [\(5.5\)](#), dry friction passing through a transition between static and kinetic regimes near zero velocity that neither a viscous nor a constant Coulomb term describes [\[23\]](#). The retained data therefore spans 63 rad s^{-1} to 94 rad s^{-1} in magnitude, and the results are shown in [Figure 5.4](#).

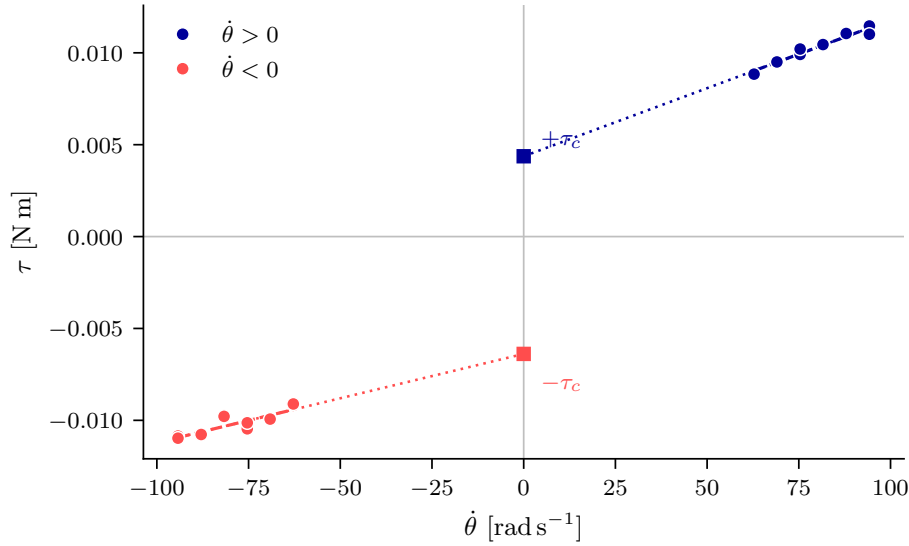


Figure 5.4: Steady-state torque against steady-state angular velocity, in both directions of rotation, with the regression of each branch.

The two branches give

$$\begin{aligned} \dot{\theta} > 0 : \quad b_m &= 7.41(70) \times 10^{-5} \text{ N m s rad}^{-1}, & \tau_c &= 4.37(55) \times 10^{-3} \text{ N m}, & R^2 &= 0.952, \\ \dot{\theta} < 0 : \quad b_m &= 4.83(110) \times 10^{-5} \text{ N m s rad}^{-1}, & \tau_c &= 6.39(88) \times 10^{-3} \text{ N m}, & R^2 &= 0.764. \end{aligned} \quad (5.6)$$

The two coefficients of determination differ enough to deserve a comment, and the difference does not mean that one branch is worse described by the model than the other. The negative branch has a flatter slope, so the torque varies less over the same velocity range, while the residual scatter is comparable in both. A coefficient of determination measures the fraction of variance the model explains, and it falls when the variance to explain is smaller, whatever the quality of the fit. The two branches agree to within their own uncertainties on both parameters, which is the comparison that matters here.

The Coulomb torque is nevertheless the weaker of the two estimates. The data spans no velocity below 60 rad s^{-1} , so the intercept is an extrapolation well outside the measured range, and its uncertainty is correspondingly larger than that of the slope.

Inertia from free deceleration

The inertia is identified by bringing the assembly to a steady velocity and then removing the torque, letting it coast to rest. With $\tau = 0$, (5.4) becomes

$$I_h \ddot{\theta} = -\tau_c \text{sign}(\dot{\theta}) - b_m \dot{\theta}. \quad (5.7)$$

This route is preferred to fitting a time constant on a step response, which would give I_h as the product of b_m with the mechanical time constant. A small b_m makes that product ill-conditioned, whereas a free deceleration measures the resistance directly, without dividing by a quantity close to zero.

Dividing by I_h gives an affine relation between the deceleration and the velocity, again with a sign depending on the direction, so the two directions are fitted separately. The common slope gives b_m/I_h and the intercepts give τ_c/I_h , from which I_h follows since b_m and τ_c are known. The angular acceleration is obtained by differentiating the recorded velocity, and the low-speed portion of each

run is excluded for the reason given above, where the friction law no longer holds and the numerical derivative becomes unreliable.

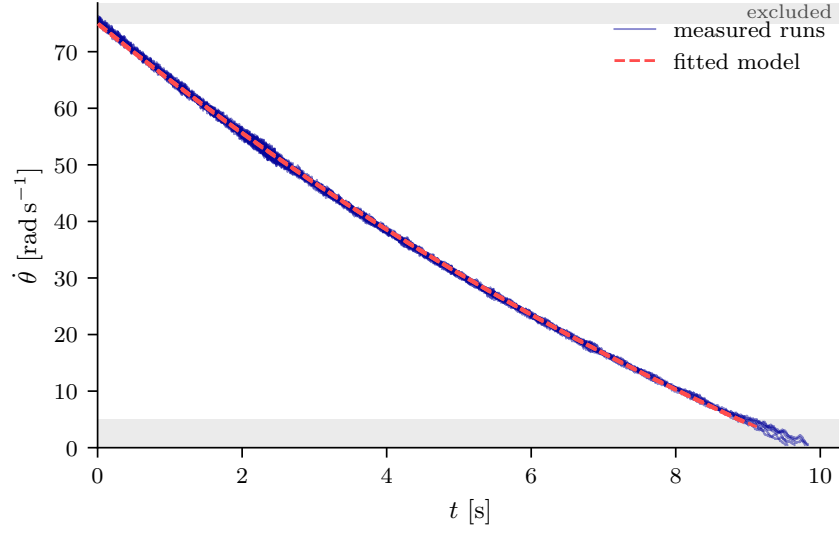


Figure 5.5: Free deceleration of the rotating assembly, with the regressions performed over the valid portion of each run.

Results

The values retained are the average of the two directions, collected in [Table 5.1](#). Two of them enter the model directly, I_h in the inertia matrix of every mode and b_m in the damping matrix. The third does not: consistently with the modelling choices of the previous chapter (see [section 3](#)), the Coulomb term is left out of the state-space matrices used for control design, since it is not smooth and cannot be represented as a linear contribution around an operating point. Its magnitude is nevertheless worth knowing, and it is invoked again when the residual errors of the following experiments are interpreted.

Quantity	Symbol	Value
Inertia of the rotating assembly	I_h	$1.12 \times 10^{-3} \text{ kg m}^2$
Viscous friction coefficient	b_m	$6.12 \times 10^{-5} \text{ N m s rad}^{-1}$
Coulomb torque	τ_c	$5.38 \times 10^{-3} \text{ N m}$

Table 5.1: Parameters identified in E1, averaged over the two directions of rotation.

5.4 E2 — Rolling friction of the ball

The second experiment identifies an equivalent ball-hoop friction coefficient b_b from the decay of a free oscillation. The coefficient is not measured directly, it is inferred from the measured amplitude envelope through the reduced ball model.

The hoop is blocked, so that its motion does not contribute to the measured dynamics. The ball is released from rest at approximately $\psi_0 = 50^\circ$, and its trajectory is recorded by the camera until the oscillation has decayed. About ten trials are performed on each hoop, and [Figure 5.6](#) shows those recorded on the outer one.

One trial on the outer hoop is excluded. The circle fitted to its trajectory returns a radius of 83.2 px against about 190 px for every other trial, and its detection rate falls to 91 % against 99 to 100 %

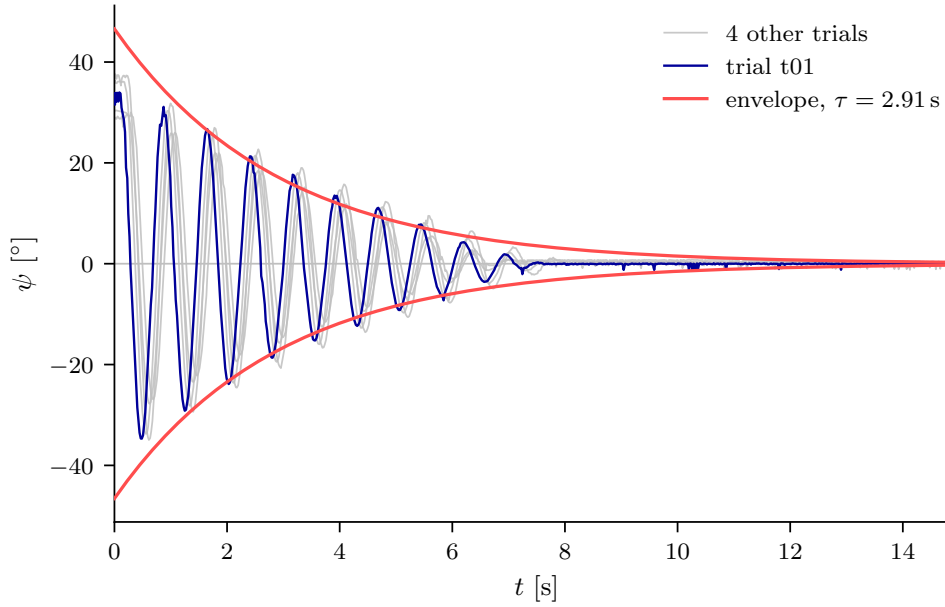


Figure 5.6: Free response of the ball released from several starting angles on the outer hoop, superposed over the trials.

elsewhere. Both indicate that the detector followed something other than the ball, so the trial carries no information about its motion.

Decay model

Successive extrema are extracted from the recorded $\psi(t)$, and the way their amplitude decays identifies the dissipation. Two candidate models are compared. A viscous term produces a geometric decay, each amplitude being a fixed fraction of the previous one, whereas a Coulomb term produces a linear decay, each oscillation removing the same amount of amplitude [23]. The two are distinguished by which of the two shapes fits the measured envelope.

The amplitude of an extremum is defined relative to the equilibrium angle, $A_k = |\psi(t_k) - \psi_{eq}|$, with ψ_{eq} estimated on the final part of each trial. This precaution matters for the smallest extrema, where a residual offset of the camera could otherwise be counted as a physical oscillation.

For a viscous decay, the logarithmic decrement and the damping ratio follow from the successive amplitudes. Two consecutive extrema being separated by half a period,

$$\delta = \frac{2}{n-1} \sum_{k=1}^{n-1} \ln\left(\frac{A_k}{A_{k+1}}\right), \quad \zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}. \quad (5.8)$$

The exponential envelope gives the lower fitting error on every trial, over the whole amplitude range covered. The dissipation is therefore compatible with a viscous model, which is the one the equations of motion carry.

Friction coefficient

With the hoop blocked, the linearised ball equation reduces to a standard second-order system,

$$\ddot{\psi} + 2\zeta\omega_0\dot{\psi} + \omega_0^2\psi = 0, \quad (5.9)$$

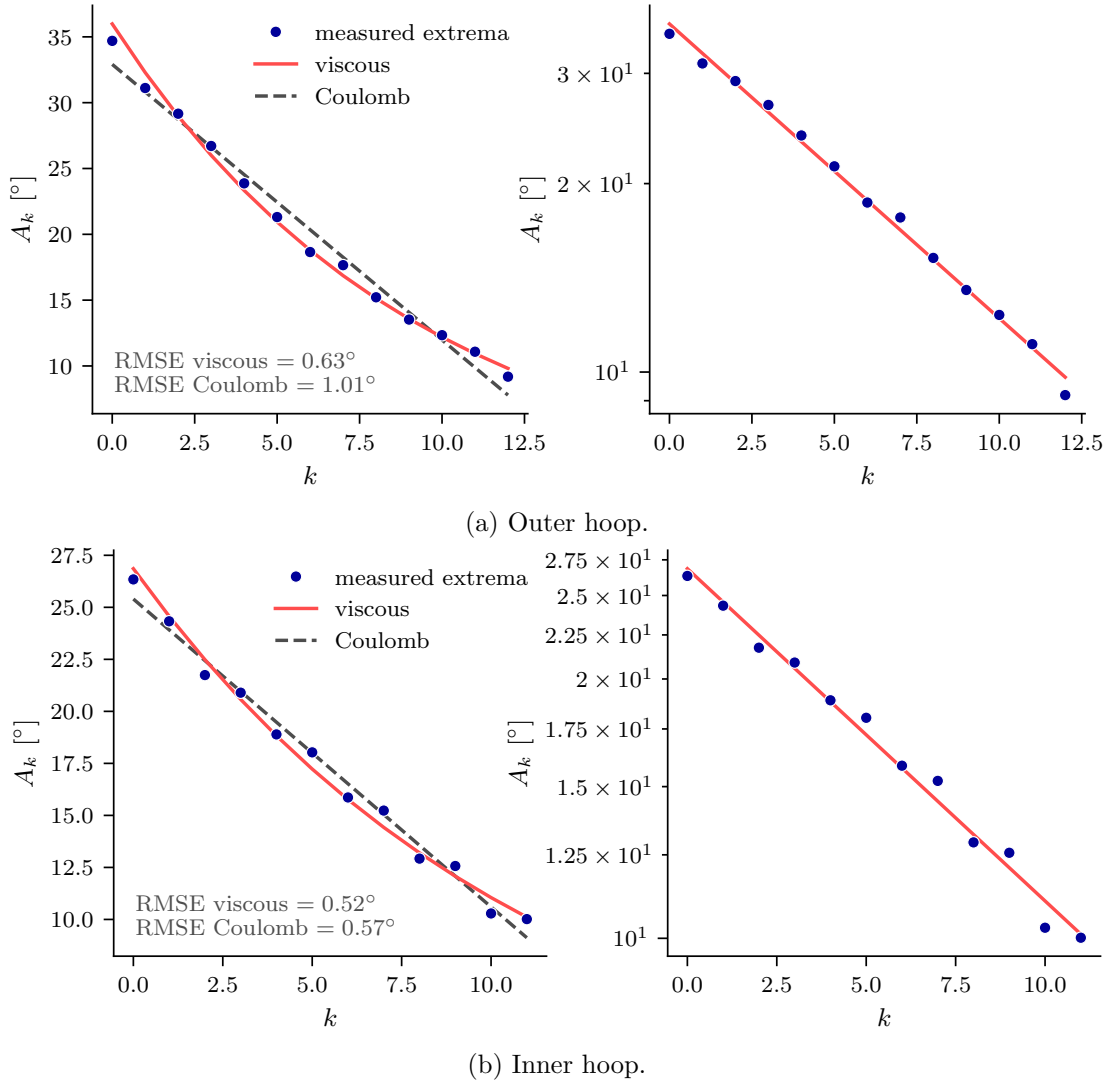


Figure 5.7: Extrema of a single trial and the two candidate envelopes fitted to them. The left panels are in linear scale, the right ones in logarithmic scale, where a viscous decay appear as a straight line.

whose coefficients are those of the corrected two-rail geometry, with R_{eff} , r_{roll} and k in place of the idealised quantities. The damping referred to the coordinate ψ is then $b_\psi = 2\zeta\omega_0 m_b k R_{eff}^2$, and referring it to the spin of the ball, which is the coordinate the model uses, accounts for the difference of lever arm,

$$b_b = b_\psi \left(\frac{r_{roll}}{R_{eff}} \right)^2 = 6.484 \times 10^{-6} \text{ N m s rad}^{-1}. \quad (5.10)$$

Geometry check

The same trials give the natural frequency of the oscillation, 1.311 Hz averaged over the campaign. The two-rail model predicts $f_0 = \frac{1}{2\pi} \sqrt{g/(kR_{eff})} = 1.32 \text{ Hz}$, within 1 % of the measurement. The idealised single-contact model, with $k = 1.4$ and $R_{eff} = R_o - R_b$, predicts 1.24 Hz, which is 5 % below.

The correction derived earlier is therefore not cosmetic. Its effect exceeds the dispersion between trials by a wide margin, so the agreement it produces is a genuine validation of the contact geometry rather than a coincidence of two adjustable quantities.

5.5 E3 — Model validation under prescribed hoop motion

Experiments E1 and E2 identify parameters from setups designed to isolate one part of the dynamics at a time. Experiment E3 asks a different question: whether the complete model, with the parameters those experiments produced, reproduces a trajectory it was not fitted to.

Protocol

The hoop is driven in position control. The ball starts at rest near $\psi = 0$ in one of the two hoops, and after half a second of baseline a position change of 45° is commanded. The ball trajectory is recorded by the camera.

Under position control, the inner loop of the driver determines the actual motion of the hoop, which is not something the simulation predicts. The measured $\theta(t)$ and $\dot{\theta}(t)$ are therefore supplied to the ball subsystem as an exogenous input, and $\psi(t)$ and $\dot{\psi}(t)$ are the only integrated variables. What is validated is the motion of the ball, decoupled from the identification of the motor.

Several trials are recorded. The camera and encoder channels are timestamped independently, and the encoder trajectory is interpolated onto the camera time grid. The latency of 19.8 ms identified during the commissioning is taken into account when aligning the two, and all derivatives are obtained through the same filtering procedure for every trial.

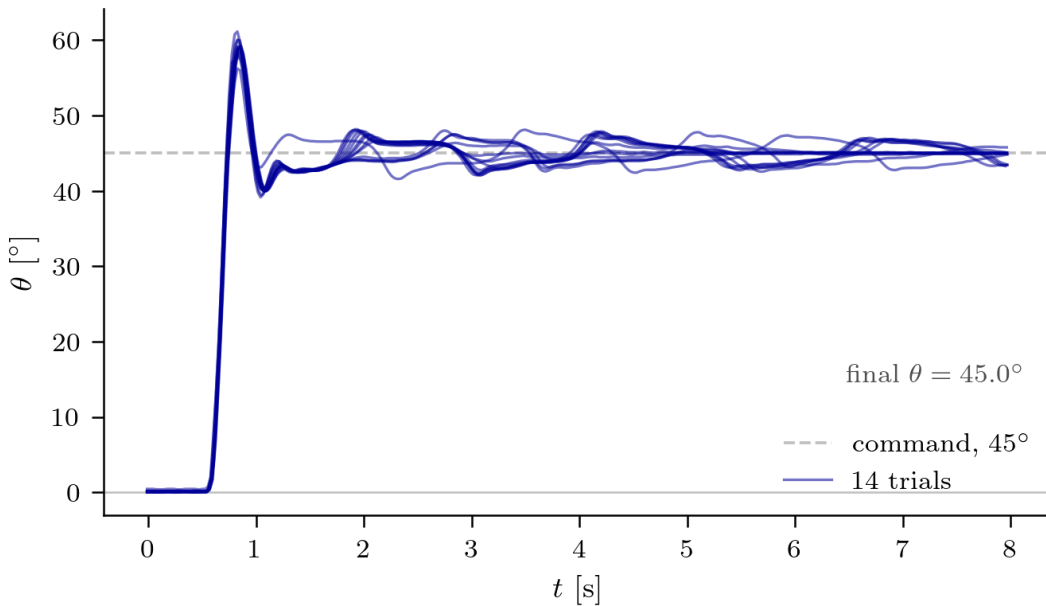


Figure 5.8: Measured hoop position over the trials of the campaign.

The detection rate is 100 % over the campaign. The effective camera sampling rate is 41.4 Hz, below the nominal 50 Hz, which remains sufficient here since the dominant ball dynamics occur near 1.3 Hz. The hoop reaches a final position of about 43.5° rather than the commanded 45° . This does not affect the validation, since it is the measured trajectory and not the command that is supplied to the simulation, but it is reported because the two are not identical.

Criteria

Two quantities are compared. The first is the root mean square error between the measured and

simulated ball position, evaluated after the start of the step,

$$\text{RMSE}_i = \sqrt{\frac{1}{N_i} \sum_{k=1}^{N_i} e_i(t_k)^2}, \quad e_i(t_k) = \psi_i(t_k) - \psi_{sim}(t_k), \quad (5.11)$$

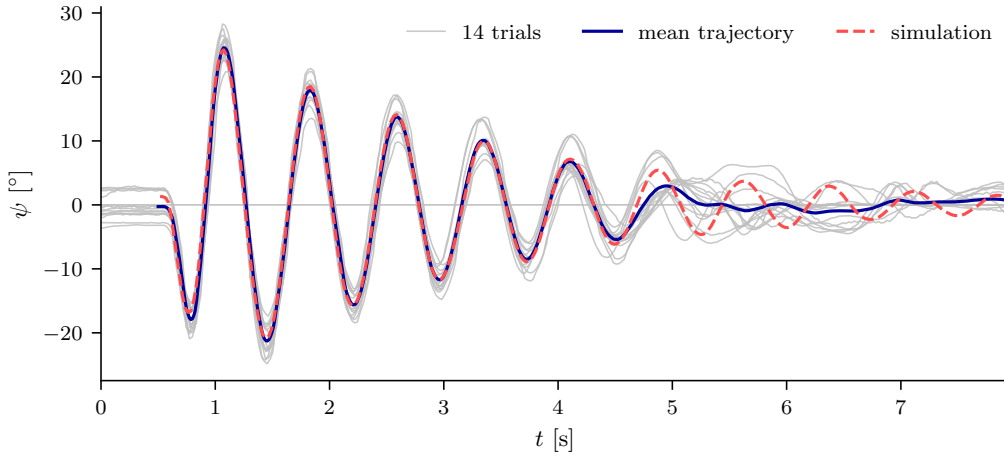
where i is the trial index and the simulated trajectory is interpolated at the camera timestamps. It is reported for each trial as well as for the mean trajectory, so that averaging cannot hide a lack of repeatability between trials.

The second is the relative error on the oscillation frequency, estimated from the intervals between successive extrema of ψ ,

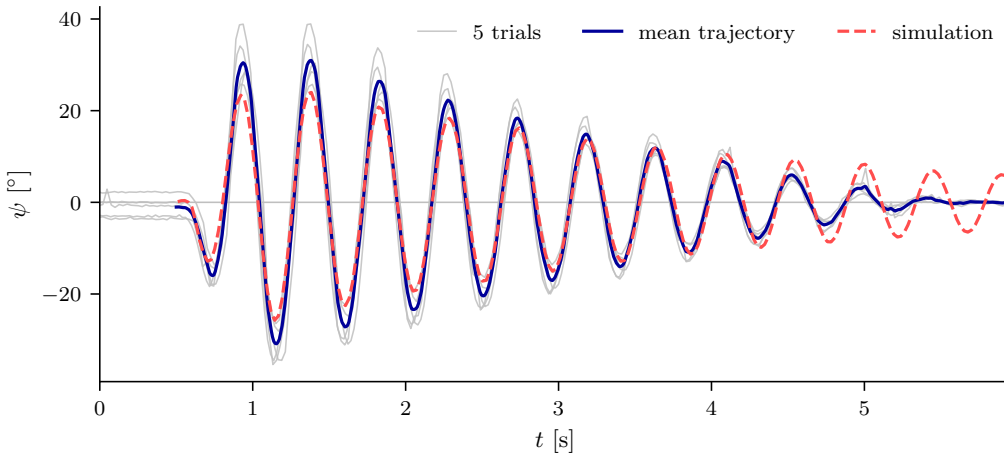
$$\varepsilon_f = \frac{|f_{sim} - f_{meas}|}{f_{meas}}. \quad (5.12)$$

The two measure different things. The first bounds the overall discrepancy, and the second isolates an error on the effective geometry, since the frequency depends on kR_{eff} alone and not on the friction.

The model is accepted if the root mean square error stays below 3° on every trial and the relative frequency error below 5 %.



(a) Outer hoop.



(b) Inner hoop.

Figure 5.9: Measured ball position over the trials, with the mean trajectory and the simulated one superposed.

Results

The campaign gives

$$\text{RMSE} = 2.47^\circ, \quad \varepsilon_f = 3.46 \%,$$

so both criteria are satisfied. The residual error is of the same order as the uncertainty on ψ established during the commissioning, which places it at the level of what the measurement itself can resolve.

Three effects can account for what remains. The alignment between the two timestamp channels is exact only to within one camera period. The friction terms left out of the model, the Coulomb torque of E1 among them, act on the trajectory without appearing in the simulation. And the effective geometry, however carefully derived, rests on two measured dimensions of the tracks whose own uncertainty propagates into R_{eff} and k .

5.6 Summary

The parameter identification is now done, and every major parameter used in the mathematical model has been assigned to a value corresponding to the real prototype. These parameters as well as the simulation itself are proven to be functional, with an error of less than 5% in terms of frequency as well as a RMSE under 3° . [Table 5.2](#) contains the same parameters as the ones introduced in the mathematical model, with their associated value.

Symbol	Description	Value	Unit
g	Gravitational acceleration	9.81	m/s ²
R_b	Ball radius	$12.5 \cdot 10^{-3}$	m
m_b	Ball mass	$67 \cdot 10^{-3}$	kg
I_b	Ball moment of inertia ($\frac{2}{5}m_b R_b^2$ for a solid sphere)	$4.19 \cdot 10^{-6}$	kg·m ²
b_b	Ball rolling friction coefficient	$6.48 \cdot 10^{-6}$	N·m·s/rad
R_o	Outer hoop inner radius	$112.5 \cdot 10^{-3}$	m
R_i	Inner hoop mid-radius	$50 \cdot 10^{-3}$	m
t_i	Inner hoop wall thickness	$11 \cdot 10^{-3}$	m
$\Delta\psi_{\text{hole}}$	Angular width of the inner hoop opening	$\pi/2$	rad
ψ_{hole}	Angular position of the opening centre	π	rad
I_h	Total moment of inertia of the rotating assembly	$1.12 \cdot 10^{-3}$	kg·m ²
b_m	Motor viscous friction coefficient	$6.12 \cdot 10^{-5}$	N·m·s/rad

Table 5.2: Physical parameters of the ball-and-hoop system.

Some elements have been adapted in this section to take the complex geometry and the sensors used in reality. These are summarised in [Table 5.3](#)

Symbol	Description	Value	Unit
r_{roll}	Rolling radius of the ball on the tracks	$10.64 \cdot 10^{-3}$	m
k	Inertia factor	1.552	[-]
d_{cam}	Distance between the camera and the hoop center	$330 \cdot 10^{-3}$	m
λ	Scale factor of the camera	$4.32 \cdot 10^{-4}$	m / px

Table 5.3: Physical parameters of the ball-and-hoop system, additions after component selection and assembly.

6 Control Design

The four tasks introduced in [section 4](#) have been stated so far as physical objectives. The system itself has been modelled (see [section 3](#)), built and identified (see [section 5](#)). This section formalises each task as a control problem, selects a method to solve it, and reports what was obtained in simulation and on the prototype.

The tasks are ordered by increasing difficulty, and so are the methods. Rather than selecting a single controller and applying it everywhere, each task is solved with the simplest tool that remains valid for it. When that tool reaches its limit, the limitation is identified explicitly, and the next task introduces the generalisation that removes it. The chapter is therefore built as a progression, in which every method contains the previous one as a particular case.

6.1 Common Framework

A single mathematical framework is used to express all four tasks. Each scenario is written as an optimal control problem, with a cost J_k to be minimised. This cost contains two terms: a running cost L_k , evaluated at every instant between the initial time t_0 and the final time t_f , and a terminal cost Φ_k , evaluated once at the end of the manoeuvre. This is the Bolza form of an optimal control problem [\[39\]](#).

$$\mathcal{P}_k : \min_{u(\cdot)} J_k = \int_{t_0}^{t_f} L_k(e(t), u(t)) dt + \Phi_k(e(t_f)) \quad (6.1)$$

subject to

- the dynamics of the active mode, $\dot{x} = f_k(x, u)$ (see [section 3](#)),
- the initial condition $x(t_0) = x_0$,
- the terminal condition $x(t_f) = x_f$, imposed when applicable.
- the path constraints $g_k(x(t), u(t)) \leq 0$ for all t ,

where $e(t) = x(t) - x_{ref}(t)$ is the state tracking error.

Writing the four tasks in this common form makes their differences explicit and finite in number. Two problems differ only through the dynamics f_k retained, the length of the horizon, the terms appearing in L_k , and the path constraints g_k declared active. The variety of methods presented in this section follows from those choices rather than from unrelated designs. For T1 and T2, the resulting problem is solved analytically through the Riccati equation. Explicit path constraints are introduced for T3 and T4.

The commanded quantity u is the angular acceleration of the hoops, as established with the model (see [section 3](#)), and every controller below produces that quantity. What the driver receives, and how the acceleration is converted into a torque, is the object of the next section.

6.2 Time-Scale Separation

All designs treat the acceleration as if it were applied to the hoops instantaneously. In reality, the command is sent to the driver, which realises it through its own internal control loop. The inner loop performances are tested and the gains are tuned until the response is quick enough, with $\tau \approx 15\text{ms}$ to consider the time-scale separation valid. This process is described in [Appendix C](#).

6.3 Task T1 — Ball Stabilisation

6.3.1 Problem Statement

The objective of task T1 is to hold the ball at a reference angular position ψ_{ref} within a given rolling mode, in the presence of small perturbations. This is a regulation problem, the simplest of the four, where the reference is a fixed point.

Two candidates present themselves naturally for such a task. A Proportional-Integral-Derivative (PID) controller can be tuned on the input-output behaviour of the system, and a Linear Quadratic Regulator (LQR) can be designed from the state-space model (see [section 2](#)). The LQR is retained here for three reasons:

1. It uses a feedback on the four states with a single input, which a PID structure does not.
2. It comes with stability margins guaranteed by construction rather than obtained by tuning [\[19\]](#).
3. It is a particular case of the time-varying regulator required by task T3 (see [subsection 6.5](#)) and is used in one of the phases of task T4 (see [subsection 6.6](#)).

Solving T1 with an LQR therefore introduces, at the lowest possible level of difficulty, the tools that the later tasks will need in full.

The task is defined from $t_0 = 0$ s over an infinite horizon. This expresses that the controller is expected to hold the ball indefinitely, rather than to complete a manoeuvre by a deadline [\[19\]](#). The running cost is quadratic, weighted by two matrices $Q \succeq 0$ and $R > 0$ that encode the relative price paid for state deviations and for control effort,

$$L_1 = e(t)^\top Q e(t) + R u(t)^2. \quad (6.2)$$

Since the horizon is infinite and the target is a fixed point, no terminal cost is required and $\Phi_1 = 0$. The running cost penalises any residual deviation already.

The quantitative targets defining success are a settling time $t_s \leq 3$ s at the 5 % criterion, an overshoot $M_p \leq 20$ %, a zero steady-state error at the operating point, and stability margins satisfying $GM \geq 6$ dB and $PM \geq 45^\circ$.

The margin requirement is crucial in this work, and could be discussed a bit more precisely. An infinite-horizon LQR with full state feedback possesses an infinite gain margin at the plant input, a gain reduction margin of 6 dB and a phase margin of at least 60° [\[19\]](#). In other words, the specifications should be met by construction for the ideal design.

These guarantees hold if the true state is available instantaneously and if the loop is analysed at the plant input. However, they do not survive the insertion of an observer between measurement and feedback [\[40\]](#) nor the transport delay of the sensing chain. The margins must therefore be verified on the implemented loop rather than invoked a priori. The results below show that this precaution was not purely academic.

6.3.2 Equilibria and Linearisation

An equilibrium state of the ball-hoop system is a pair (x_{eq}, u_{eq}) satisfying $f(x_{eq}, u_{eq}) = 0$, that is a configuration in which all accelerations vanish and which the system would hold indefinitely if left undisturbed [41]. Evaluating the ball equation of motion at zero velocity gives

$$M_{21} u_{eq} + m_b g R_{eff} \sin \psi_{eq} = 0. \quad (6.3)$$

This equation admits three equilibria with $u_{eq} = 0$ and $\sin \psi_{eq} = 0$, the latter giving $\psi_{eq} \in \{0, \pi\}$. These correspond to the physical configurations in which the ball rests at the bottom or at the top of a hoop, and are collected in Table 6.1, using the same mode names as in the model (see section 3).

Mode	Surface	ψ_{eq}	Physical configuration	Stability
q_1	Outer hoop	0	Ball at the bottom of the outer hoop	Stable
q_3	Inner hoop, concave	0	Ball at the bottom of the inner hoop	Stable
q_4	Inner hoop, convex	π	Ball on top of the inner hoop	Unstable

Table 6.1: Equilibria of the ball-hoop system in each rolling mode.

The hoop angle θ does not appear in (6.3), so the equilibria are independent of the absolute hoop orientation. This is the mathematical expression of the rotational symmetry of the problem: the hoop may be turned to any orientation without changing the equilibrium of the ball. This property is used again in subsection 6.5.

Since the system admits equilibria, it may be linearised about them. To linearise means to replace the exact equations of motion by the simplest model reproducing them correctly for small deviations. For the natural equilibria $\psi_{eq} \in \{0, \pi\}$, the first-order expansion gives $\sin(\psi_{eq} + \delta\psi) \approx \cos(\psi_{eq}) \delta\psi$, since $\sin(\psi_{eq}) = 0$. The resulting linearised gravity stiffness is therefore $k_g = m_b g R_{eff} \cos(\psi_{eq})$, with $k_g > 0$ at the bottom and $k_g < 0$ at the top. The price of this simplification is that the resulting model is, of course, trustworthy only within a neighbourhood of the equilibrium. This is true for this task, but is not anymore in subsection 6.5 and subsection 6.6 for example, where the manoeuvre deliberately leaves that neighbourhood. The linearisation is written directly in terms of the error $e = x - x_{eq}$,

$$\dot{e} = A e + B u, \quad (6.4)$$

and the matrices are obtained by differentiating the equations of motion [2],

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -C_{21}/M_{22} & -k_g/M_{22} & -C_{22}/M_{22} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -M_{21}/M_{22} \end{bmatrix}, \quad (6.5)$$

where M_{21} , M_{22} , C_{21} and C_{22} are entries of the rolling-mode matrices (3.23) evaluated at the equilibrium, and $k_g = m_b g R_{eff} \cos \psi_{eq}$ is the linearised gravity stiffness.

The sign of k_g is what indicates the stability of the equilibrium. If $\cos \psi_{eq} > 0$, the entry $-k_g/M_{22}$ is negative and behaves as a restoring spring bringing the ball back towards its equilibrium. The equilibrium is stable, and gravity does the work of the controller. If $\cos \psi_{eq} < 0$, which occurs at $\psi_{eq} = \pi$, the same term behaves as a repulsive spring accelerating the ball away from its equilibrium. The configuration is that of an inverted pendulum, unstable, and only feedback can hold it.

The structure of (6.5) also reveals the actuation architecture. The hoop states, in the first two rows, evolve independently of the ball. The ball states, in the last two rows, are driven by the hoop velocity and by the motion of the ball itself. The motor drives the hoop, and the ball responds passively to that motion. As already discussed with the properties of this system (see subsection 2.2), this is the translation of underactuation, which of course remains true with this linearised form.

With the linearisation established, the problem can be written in full,

$$\mathcal{P}_1 : \min_{u(\cdot)} J_1 = \int_0^\infty [e(t)^\top Q e(t) + R u(t)^2] dt \quad (6.6)$$

subject to $\dot{e} = Ae + Bu$, to the actuator saturation $|u| \leq u_{max}$, and with e the deviation from an equilibrium state x_{eq} . The problem admits a unique stabilising solution as long as the pair (A, B) is stabilisable and the pair $(A, Q^{1/2})$ is detectable [19]. Both conditions are verified numerically for each rolling mode, the controllability and observability, and are validated.

An element should be expressed here, because it is a weakness of the formulation: the bound $|u| \leq u_{max}$ appears among the constraints of $\mathcal{P}_1$, but the solution derived below does not enforce it. This solution is the closed-form optimum of the unconstrained problem, since this analytical resolution through the Riccati equation is possible precisely in the absence of bounds. The weight R makes large inputs expensive without making them impossible. A large initial deviation therefore produces a commanded value above u_{max} , which the driver then saturates. This means that the bound is respected only within a range of initial deviations, which is a distinct limitation from the linearisation validity. This is verified by testing the task on a small deviation and on a larger one, on the outer hoop. The same large deviation is then tested on the inner hoop.

6.3.3 Control Law

Problem $\mathcal{P}_1$, with linear dynamics and quadratic cost, is the classical Linear Quadratic Regulator problem [19]. Its optimal solution is a static linear state feedback,

$$u^*(t) = u_{eq} - K e(t), \quad (6.7)$$

where u_{eq} is the feedforward at the operating point and $K \in \mathbb{R}^{1 \times 4}$ the constant gain minimising J_1 . The law has two distinct roles, worth separating clearly.

Feedforward

At the natural equilibria, the static balance gives $u_{eq} = 0$, since no effort is needed to stay where gravity already holds the ball. If the controller is used for an off-equilibrium reference, gravity no longer balances and a constant effort is required,

$$u_{eq}(\psi_{ref}) = -\frac{m_b g R_{eff} \sin \psi_{ref}}{M_{21}}. \quad (6.8)$$

This term compensates the static gravity torque at the operating point, and thereby yields a zero steady-state error without introducing an integrator into the controller. Another option would consist in adding an integral state to cancel the static error, but the margins recalled above are a property of the LQR designed on the original plant. They do not hold for an augmented-state LQR, where the robustness of the loop would need to be re-established from scratch [40].

The trade-off of using feedforward is that it is only as accurate as the model is. It can cancel the static error exactly if m_b , R_{eff} and M_{21} are exact, and leaves a residual bias in proportion to their error. Furthermore, (6.8) grows with $\sin \psi_{ref}$ while the linearisation degrades with the distance to the equilibrium. Far from it, the required input may exceed u_{max} , even if the linear model is still valid.

Feedback

The optimal gain follows from the positive-definite solution P of the algebraic Riccati equation (see section 2), through $K = R^{-1}B^\top P$. The matrices Q and R are the only design freedom left, and their choice is guided by the rule of Bryson [42], which normalises each weight by the square of the largest deviation deemed acceptable for the corresponding quantity,

$$Q = \text{diag}\left(\frac{1}{\theta_{max}^2}, \frac{1}{\dot{\theta}_{max}^2}, \frac{1}{\psi_{max}^2}, \frac{1}{\dot{\psi}_{max}^2}\right), \quad R = \frac{1}{u_{max}^2}. \quad (6.9)$$

The rule makes the weights dimensionally homogeneous and gives a good first approximation of their value. Since it is based on physical quantities, it provides an easy comparison between the different tasks. The retained values and the resulting closed-loop poles are reported in Table 6.2.

6.3.4 Results

The design is first applied to the simulation, which is extended to simulate the closed-loop, adding the sensors parameters and delays, the inner loop, the linearisation of the dynamics, the gains calculation and the control law itself. Once the simulation works, the design is transferred to the prototype.

Both are reported together, and a discussion is made on the elements that separate the simulated performance from the measured one. Unless stated otherwise, the results below concern the outer rolling mode q_1 .

Simulation

The simulation serves two purposes: verifying that the chain of linearisation, resolution and feedback is internally consistent, and obtaining a first set of gains to be transferred to the bench. The weights retained after adjustment of the starting point given by (6.9), the resulting feedback gain and the closed-loop properties it produces are collected in Table 6.2.

Quantity	Symbol	Value
State weights	Q	$[1/(10\pi)^2, 1/2^2, 1/0.1^2, 1/0.3^2]$
Input weight	R	$5 \cdot 10^{-5}$
Feedback gain	K	$[1, 4, 18, 5.5]$

Table 6.2: Retained weights for task T1.

Two recovery tests are simulated, from initial deviations of 10° and 45° . The first case verifies the specifications in the regime for which the design is intended. The second case challenges the validity of the linearisation itself, by comparing the non-linear simulation with the prediction of the linear model, giving an empirical estimate of the neighbourhood over which the design can be trusted. Since the saturation is not enforced by construction, the maximum commanded input is reported explicitly and compared with u_{max} .

Transfer to the prototype

Applying these gains to the bench without more treatment did not reproduce the simulated behaviour. Two obstacles are met, and neither concerns the gain values themselves: the first concerns a dependency of stability on the loop frequency, and the second one concerns the gains applied on the inner rolling mode.

Frequency dependency: With identical gains, the closed loop was found to be stable at 50 Hz but divergent at 36 Hz, as shown in Figure 6.1. Since nothing changed in the design between the two runs, the instability is not attributed to the gains, but rather to the latency of the chain, dominated by the camera.

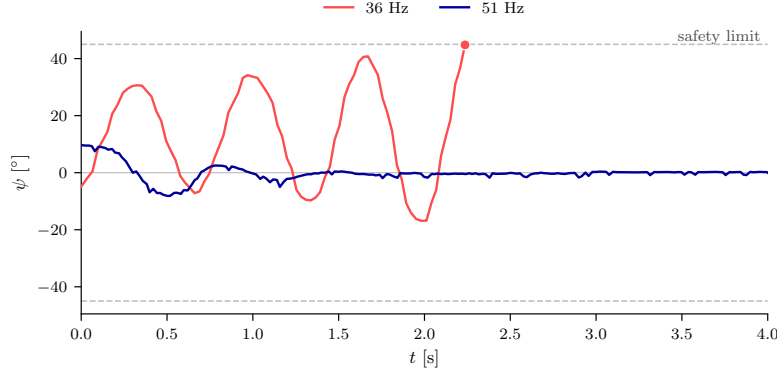


Figure 6.1: Measured ball position at two loop frequencies, with identical gains and from comparable initial excursions.

This result shows how important sampling frequency can be on the stability of a loop, independently from any choice of gains. The closed-loop, during the first tests, was not optimised at all: the USB-communication occurred at many instants during the loop, which added a delay of few ms each time, and the latency of the sensing chain, dominated by the camera, already consumes most of the phase available to the loop. The loop is therefore optimised to hold the frequency at 50 Hz for every result that follows, after some failed attempts.

The USB communication is restructured so that a single exchange with the driver is made per control period, instead of the several that were spread over the loop. The camera delay is compensated by propagating the state estimate forward over the measured latency, following the approach of the reference project [2]. Together, these two changes bring the loop from 36 Hz to the 50 Hz targeted by the design.

Gains applied to the inner rolling mode: The gains used for the inner rolling mode have been computed using the natural frequency of the outer hoop instead of the inner one, the value having been carried over rather than recomputed. The correct value for the inner hoop is $f_n = 2.2157$ Hz, against $f_n = 1.3144$ Hz for the outer one. This of course has an influence on the first attempts made on the inner hoop. The final runs correct this simply by re-evaluating (6.5) for this mode rather than re-used.

Performance

With the corrections applied, the regulator is run on the prototype. **Figure 6.2** compares the measured ball position with its reference, and the commanded input with the value actually delivered by the driver. The performance obtained is compared with the specifications and with the simulated values in **Table 6.3**.

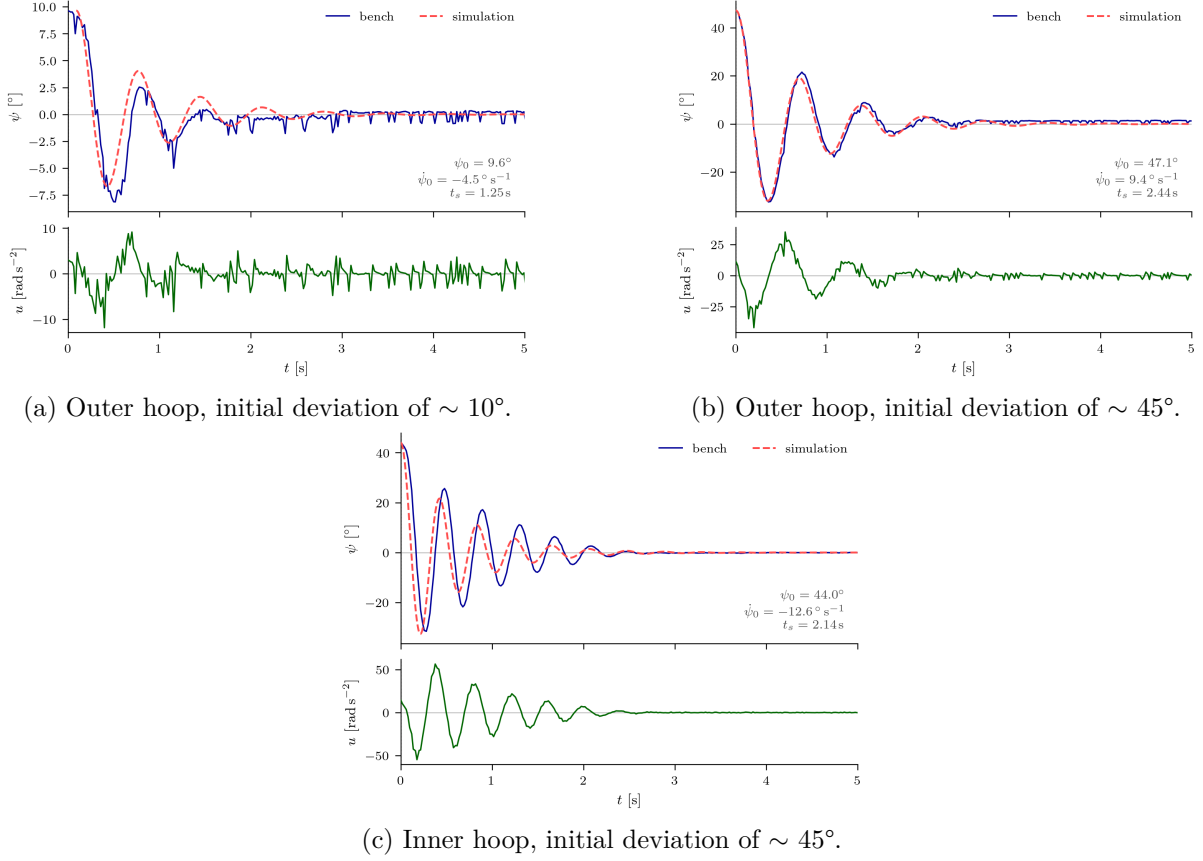


Figure 6.2: Comparison between experimental LQR recovery trajectories and aligned simulations from the same initial conditions.

Criterion	Specification	Simulation	Bench
Settling time t_s	≤ 3 s	2.13	2.18
Overshoot M_p	≤ 20 %	69.4	84.3
Steady-state error	0	0°	0.18°
Peak input $\max u $	$\leq u_{max}$	-	11.8

Table 6.3: Performance of the regulator against the specifications of task T1, for the outer hoop.

Three of the four criteria are met. The settling time reaches 2.18 s on the bench against 2.13 s in simulation, inside the 3 s specification, and the peak input stays below the actuator bound. The steady-state error of 0.18° falls below the 0.33° uncertainty of the vision chain, so it cannot be distinguished from the resolution of the measurement itself. The overshoot however fails, at 84.3 % against the 20 % required and with a simulation already at 69.4 %. The design is the first cause: the weights favour a short settling time over a damped return, and the latency of the sensing chain widens the gap. The cost of this overshoot is however low, since the ball stays on tracks and comes back.

The regulator however holds the ball at both equilibria and recovers from deviations beyond the linearisation neighbourhood.

6.4 Task T2 — Reference Tracking

6.4.1 Problem Statement

The second task extends the first to a time-varying reference $\psi_{ref}(t)$, within the same rolling modes. The reference is assumed to remain close enough to the equilibrium of T1 for the linearised dynamics (6.4) to stay valid, with the same matrices A and B . The reference state contains a prescribed ball trajectory. The hoop angle is left free, the ball position and velocity being the only quantities that must track a reference. With the same error variable and the same quadratic cost as before, the problem keeps the structure of $\mathcal{P}_1$,

$$\mathcal{P}_2 : \quad \min_{u(\cdot)} J_2 = \int_0^\infty [e(t)^\top Q e(t) + R u(t)^2] dt \quad (6.10)$$

subject to $\dot{x} = Ax + Bu$ and $|u| \leq u_{max}$. The only difference with $\mathcal{P}_1$ lies in the reference, which is no longer constant.

6.4.2 Control Law

Without modification, the problem reuses the gain K and the feedforward u_{eq} designed for T1, applied this time to a non-constant error,

$$u(t) = u_{eq} - K e(t). \quad (6.11)$$

Substituting (6.11) into the linearised dynamics and using $\dot{e} = \dot{x} - \dot{x}_{ref}$ gives

$$\dot{e}(t) = (A - BK) e(t) + w(t), \quad w(t) = A x_{ref}(t) + B u_{eq} - \dot{x}_{ref}(t). \quad (6.12)$$

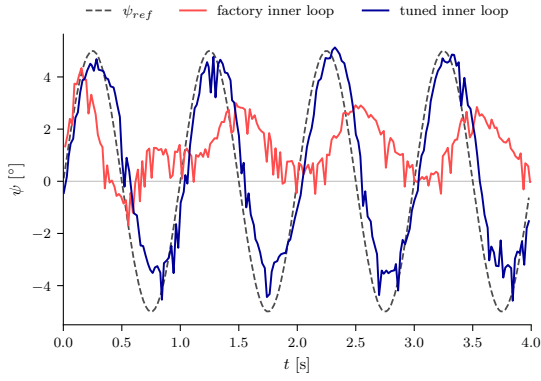
The interpretation of (6.12) is what makes this extension clear. The matrix $A - BK$ is the same as in T1, so the way the loop rejects the error itself is unchanged. What is new is $w(t)$. This term acts as a forcing input on the error itself, measuring the mismatch between the reference motion and the motion the plant would spontaneously produce at this operating point. When the reference is constant, the forcing term vanishes and (6.12) reduces exactly to the closed loop of T1. T2 is a generalisation of T1, with a controller permanently chasing a target it never completely reaches.

Since $A - BK$ is stable by construction, the error remains bounded for any bounded forcing, and its magnitude in steady state is governed by the magnitude of $w(t)$ [43]. Not every reference is admissible. The ball is never directly actuated, and the only way to move it is to move the hoop first. A reference that asks the ball to change direction faster than the loop can rotate the hoop and let the contact make the ball change direction can therefore not be followed. The admissible references are those whose rate of change remains small compared with the bandwidth that $A - BK$ imposes, covering smoothed steps, ramps, and low-frequency sinusoids.

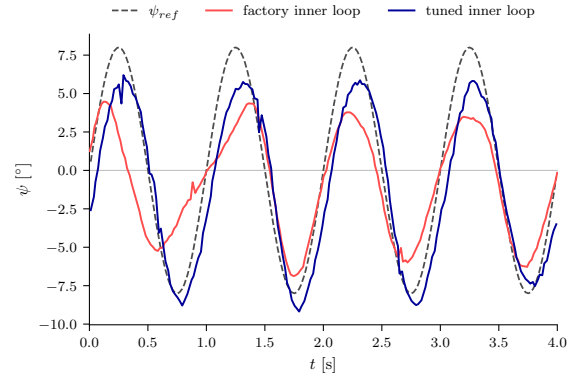
6.4.3 Results

The first results using this task were obtained with the factory gains of the inner hoop, before the tuning was actually performed. This task is a very good way of showing how the results have been improved with the inner loop improvement, shown in Figure 6.3.

Several sinusoidal references are tested, sweeping several amplitudes. For each pair (A_{ref}, f_{ref}) , the tracking error is evaluated after the initial transient. The tracking performance is characterised from these two sweeping tests.



(a) Comparison on a 5° sinusoidal 1Hz reference.

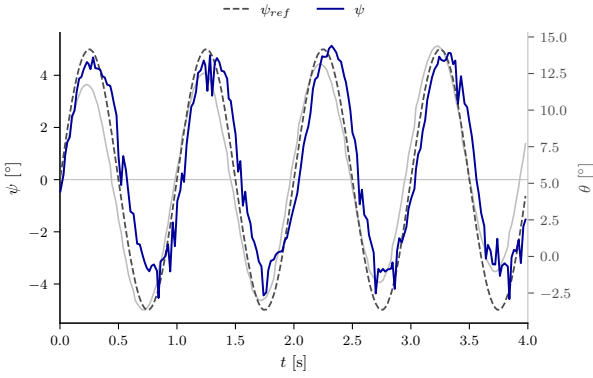


(b) Comparison on a 8° sinusoidal 1Hz reference.

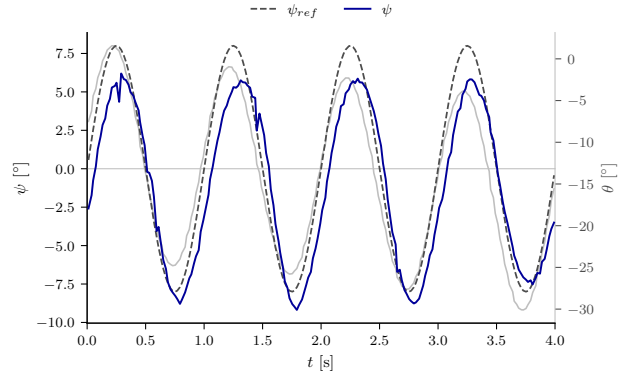
Figure 6.3: Comparison between the tracking performance with the factory gains of the inner loop (red) and the tuned gain (blue) values.

Tracking at different amplitudes

The reference is followed at both amplitudes, with a visible phase lag and an attenuation that grows



(a) Outer hoop, sinusoidal reference of 5° at 1 Hz.



(b) Outer hoop, sinusoidal reference of 8° at 1 Hz.

Figure 6.4: Measured ball position against its reference, with the hoop angle commanded by the regulator shown on the right axis.

with the amplitude. The lag is expected: the ball is reached only through the hoop, and the loop must therefore build the hoop motion before the contact transmits it. The right axis of each plot shows the price paid for that: the hoop travels an excursion several times larger than the one asked of the ball. It also grows faster than the reference itself. This is the underactuation of the system read on a measurement: the actuated coordinate is the one that has to move, and the controlled one follows.

Performance

Two conclusions follow from these tests. The first concerns the inner loop. With the factory gains, the tracking is qualitatively wrong: the ball barely follows the reference at 5° and lags at 8°. The outer gains are identical in both runs, which excludes the design of T2 for this difference. With the factory gains, the inner loop has $\tau = 196$ ms, a fifth of the reference period, so the hoop no longer reaches the commanded acceleration before the command has changed and the time-scale separation no longer holds.

The second concerns the reach of the method. T2 uses the T1 regulator and reproduces a reference that stays within a few degrees of the equilibrium. Nothing in this design would allow the ball to be brought far from it. The tasks T3 and T4 leave this neighbourhood entirely, which motivates the use of a more complex method in the following sections.

6.5 Task T3 — Loop-the-Loop

Task T3 consists in making the ball complete one full revolution of the outer hoop while remaining in contact with it. This manoeuvre starts and ends at rest at the bottom of the hoop. Unlike the regulation and tracking tasks considered previously, the ball must travel a large excursion and pass over the top of the hoop.

The top of the hoop introduces a specific physical constraint: the ball is not attached to the track, so contact can only be maintained if the track provides a compressive normal force on the ball. The contact is maintained only while $N \geq 0$ (see [section 3](#)). At the top of the hoop, where $\psi = \pi$, this condition becomes $R_{\text{eff}}\dot{\psi}^2|_{\psi=\pi} \geq g$. The ball must therefore reach the top with sufficient velocity to generate the required centripetal acceleration. In consequence, the looping manoeuvre is a constrained energy-transfer problem, in which the actuator must build up velocity while preserving contact.

Tasks T1 and T2 cannot address this objective directly, since they are based on a linearisation around an equilibrium, and are designed for small deviations or slow reference trajectories. A complete revolution of the ball clearly leaves this neighbourhood. Task T3 requires a different approach: the full nonlinear rolling dynamics are used to generate a nominal trajectory that satisfies the actuator and contact constraints, before stabilising this trajectory by a time-varying feedback controller. This section starts by stating the problem in the same form as for the two first tasks, before treating the trajectory generation followed by the control feedback law and the obtained results.

6.5.1 Problem Statement

T3 uses the full nonlinear rolling-mode dynamics rather than their linearisation. The cost keeps the same structure of [\(6.1\)](#), with a running cost penalising the control effort only,

$$L_3(u) = u(t)^2, \quad \Phi_3 = 0, \quad (6.13)$$

so that the problem seeks the minimum-effort manoeuvre achieving the loop rather than the minimum-time one. A minimum-time formulation could drive the actuator into permanent saturation or close to its limits, leaving no control authority for the feedback. The final time is itself a decision variable, bounded within an admissible range rather than fixed, since the duration of the optimal manoeuvre is not known in advance. The problem is then

$$\mathcal{P}_3 : \min_{u(\cdot), t_f} J_3 = \int_0^{t_f} u(t)^2 dt \quad (6.14)$$

subject to

- the nonlinear dynamics $\dot{x} = f(x, u)$ of the rolling mode,
- the initial condition $x(0) = x_0$, the ball being at rest at the bottom equilibrium,
- the terminal condition on the hoop velocity $\dot{\theta}(t_f) = 0$ and on the ball state $\psi(t_f) = 2\pi$, $\dot{\psi}(t_f) = 0$,
- the actuator constraint $|u(t)| \leq u_{\max}$,
- the contact constraint $N(\psi(t), \dot{\psi}(t)) \geq \varepsilon m_b g$ for all t ,
- box constraints on the states, keeping the solver within the physically meaningful region.

The terminal condition leaves the hoop angle free, as the dynamics depend on the hoop velocity but not on the angle itself. Forcing it would shrink the feasible set. The contact constraint carries a positive margin to make sure that some robustness to model error or disturbances is kept. Increasing the margin requires a faster trajectory, and a larger control effort. Its admissible value is consequently limited by the actuator constraint.

6.5.2 Trajectory Generation

The optimal-control problem $\mathcal{P}_3$ is infinite-dimensional, since the control is a function of time. A numerical trajectory cannot be obtained by optimising a finite set of values without first introducing a transcription of the continuous problem.

Direct collocation is used here to convert $\mathcal{P}_3$ into a finite-dimensional nonlinear program. The state and input trajectories are represented by their values on a temporal mesh, while defect constraints enforce consistency with the nonlinear dynamics. The actuator and contact constraints are imposed along the discretised trajectory, and the final time is optimised together with the state and input values.

Nonlinear state-space model

For the rolling mode considered in T3, the state is $x = [\theta \ \dot{\theta} \ \psi \ \dot{\psi}]^\top$, and the input is the hoop angular acceleration $u = \ddot{\theta}$. The nonlinear dynamics used by the optimiser are

$$\dot{x} = f(x, u) = \begin{bmatrix} \dot{\theta} \\ u \\ \dot{\psi} \\ -\frac{M_{21}u + C_{21}\dot{\theta} + C_{22}\dot{\psi} + m_b g R_{\text{eff}} \sin \psi}{M_{22}} \end{bmatrix}. \quad (6.15)$$

The coefficients M_{21} , M_{22} , C_{21} , and C_{22} come from the mathematical model in rolling mode (see [section 3](#)). They are constant within the considered mode. The nonlinearity retained by the optimiser is the exact gravitational term $\sin \psi$.

Direct collocation

The time interval $[0, t_f]$ is divided into $K - 1$ intervals with K nodes. The final time, t_f , is a decision variable, so the physical interval length is $h = t_f / (K - 1)$. A Hermite–Simpson transcription is used. The state is approximated by a cubic polynomial on each interval and the input by a linear interpolation. The dynamics are enforced through the defect constraint

$$\delta_k = f(x_c, u_c) - \dot{x}_c = 0, \quad k = 1, \dots, K - 1, \quad (6.16)$$

where x_c , u_c , and $\dot{x}_c$ denote the interpolated midpoint state, input, and state derivative.

This transcription converts the continuous-time problem into a finite-dimensional nonlinear program whose decision variables are the nodal states, nodal inputs, and final time. The discretised cost is evaluated using the same Simpson quadrature as the dynamics,

$$J_{3,d} = \sum_{k=1}^{K-1} \frac{h}{6} (u_k^2 + 4u_c^2 + u_{k+1}^2). \quad (6.17)$$

The initial and terminal conditions are imposed at the first and last nodes. The input and contact constraints are imposed at every node,

$$|u_k| \leq u_{\max}, \quad N(\psi_k, \dot{\psi}_k) \geq \varepsilon m_b g, \quad k = 1, \dots, K. \quad (6.18)$$

Since the path constraints are imposed on a finite mesh, their satisfaction between nodes is not guaranteed by the nonlinear program alone. The resulting nominal trajectory is therefore re-simulated on the continuous-time dynamics with the computed input, and the contact force and actuator command are checked throughout the manoeuvre.

The final time is optimised together with the trajectory. Without an upper bound, the minimum-effort cost could be reduced by performing the manoeuvre arbitrarily slowly. Conversely, an excessively short final time would require large accelerations and would violate the actuator bound. The admissible interval $t_f \in [T_{\min}, T_{\max}]$ therefore defines the range of manoeuvre durations considered by the optimisation.

Nominal trajectory

The nonlinear program is solved using the direct-collocation transcription described above. The solution provides a nominal state trajectory $x^*(t)$, a nominal input $u^*(t)$, and an optimised final time t_f^* . In simulation, the trajectory completes one revolution, returns to the bottom equilibrium, and satisfies the imposed actuator and contact constraints at the collocation nodes, as shown in [Figure 6.5](#).

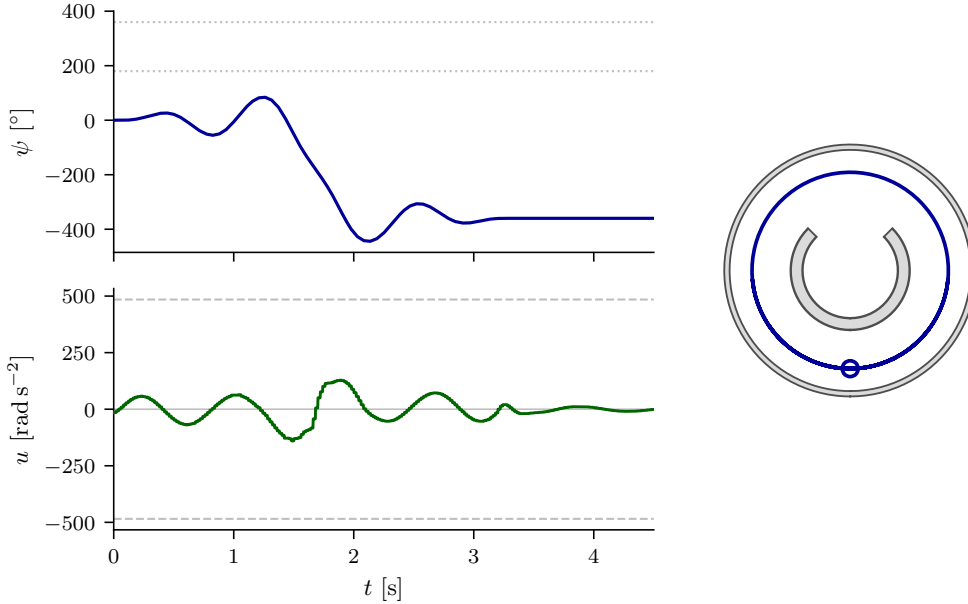


Figure 6.5: Nominal trajectory of the looping manoeuvre.

This nominal trajectory is not controlled yet. It serves as a reference that the controller must track, in the presence of initial errors, modelling uncertainties, measurement noise, or actuation delay. The tracking controller still needs to be designed.

6.5.3 Control Law

The collocation solution provides a nominal state-input trajectory $(x^*(t), u^*(t))$. Applied in open loop, this trajectory is sensitive to initial errors or model mismatch, which may be amplified near the top of the hoop. A feedback law is therefore required to track the nominal trajectory.

The dynamics are linearised along $(x^*(t), u^*(t))$ by defining $\delta x = x - x^*$ and $\delta u = u - u^*$, leading to the linear time-varying model

$$\delta \dot{x} = A(t) \delta x + B(t) \delta u, \quad A(t) = \left. \frac{\partial f}{\partial x} \right|_{x^*(t), u^*(t)}, \quad B(t) = \left. \frac{\partial f}{\partial u} \right|_{x^*(t), u^*(t)}. \quad (6.19)$$

A finite-horizon time-varying LQR is then used to compute the feedback gain. The gain is obtained by integrating the differential Riccati equation backwards from the terminal condition $S(t_f) = S_f$,

$$-\dot{S} = A^\top S + SA - SBR^{-1}B^\top S + Q, \quad K(t) = R^{-1}B(t)^\top S(t). \quad (6.20)$$

The implemented control law is

$$u(t) = u^*(t) - K(t) \delta x(t). \quad (6.21)$$

The nominal trajectory and the gains are computed offline and interpolated during the manoeuvre. The TVLQR adapts the feedback gain to the local dynamics along the trajectory, giving a locally valid feedback. However, no global guarantee exists for the nonlinear system, and its performance in the presence of state-estimation errors, delays or model mismatch must be addressed experimentally.

6.5.4 Results

The results are presented in two stages. First, the TVLQR control is applied and evaluated in simulation. The same manoeuvre is then attempted on the prototype, and the discrepancy between both is analysed.

Simulation

The manoeuvre is considered successful in the simulation, since the ball completes a full revolution with the imposed actuation and normal-force constraints, and returns to the bottom equilibrium, as shown in [Figure 6.6](#). The nonlinear program converges to a locally optimal solution satisfying the different constraints.

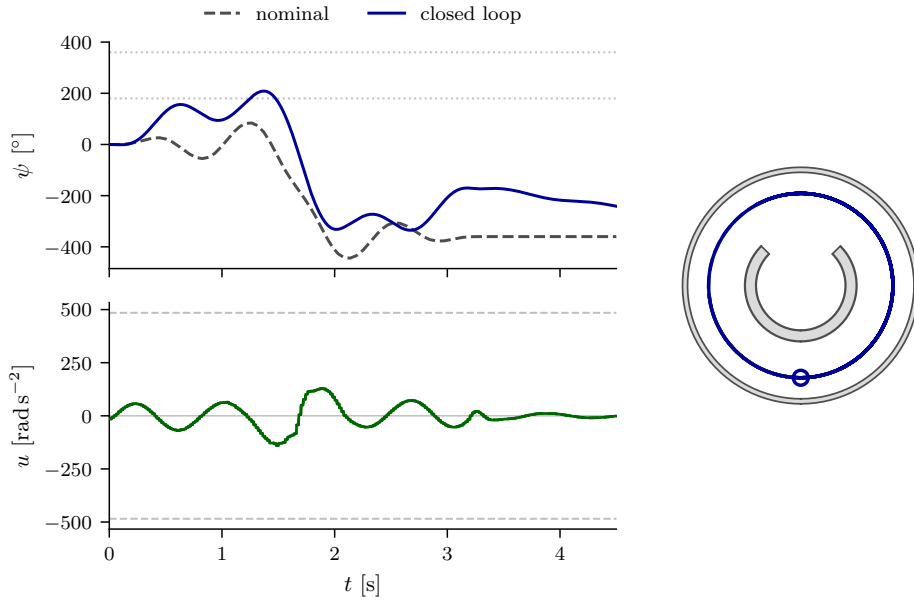


Figure 6.6: Simulation of the looping manoeuvre.

When the feedback law is applied, the simulated state remains close to the nominal trajectory and completes the looping without violating any constraint.

Bench

The same manoeuvre is applied to the physical prototype. However, the ball departs from the track at approximately 98° , before reaching the top, and falls back. This departure point is repeated across all trials, indicating that the failure is systematic rather than caused by measurement noise alone. The measured trajectories are shown in [Figure 6.7](#).

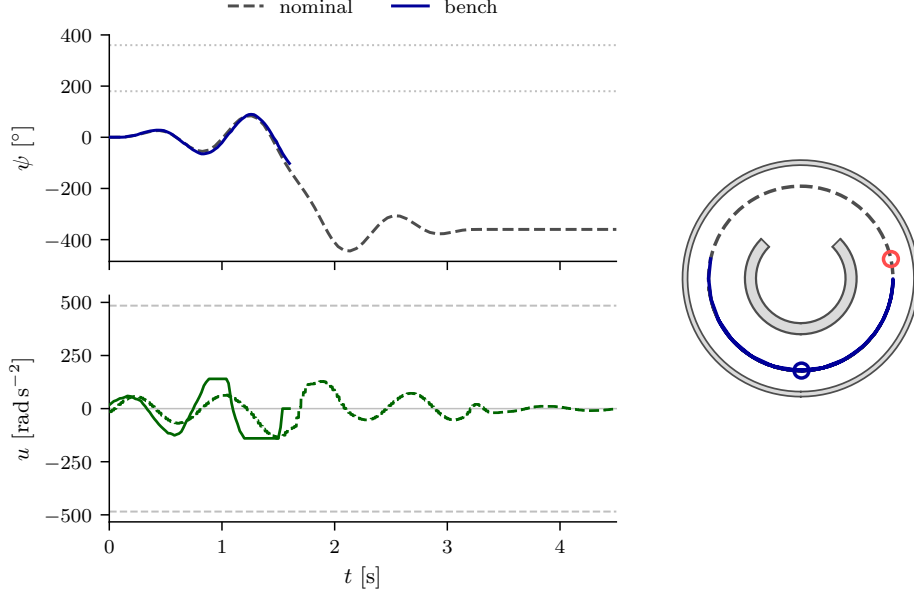


Figure 6.7: Results of the looping manoeuvre on the bench, compared to the simulation results. The simulation does the loop, while the bench fails.

The discrepancy between simulation and experiment could originate from a limitation of the reduced-order rolling model. The no-slip constraint is imposed analytically in the rolling dynamics, which eliminates any tangential contact force from the equations. Consequently, the optimisation problem has no variable through which a violation of this kind could be detected. The simulation could be improved by adding a condition on the friction coefficient required by the nominal trajectory to the optimisation problem. This would make sure that the contact force, $|F_t| \leq \mu N$, could remain non-zero during the whole looping.

The discrepancy could also be explained by the sensing and actuation delays, that reduce the bandwidth available to the feedback controller. Even if this could indeed improve the feedback, it would not remove the contact-feasibility problem, as even a perfectly tracked trajectory could require an unattainable effective friction coefficient, in this configuration.

The tangential force can be reconstructed a posteriori from the ball force balance. The corresponding minimum effective friction required to sustain the nominal trajectory is

$$\mu_{\text{req}}(\psi) = \frac{|F_t(\psi)|}{N(\psi)} \approx 3.7. \quad (6.22)$$

The effective friction available on the prototype is much lower (see [subsection 4.3](#)). In conclusion, the TVLQR successfully stabilises the nominal looping trajectory in the reduced-order simulation. However, the trajectory is not physically feasible for the prototype, because the no-slip model does not enforce contact friction.

6.6 Task T4 — Flying Ball

Task T4 is the most demanding of the four tasks. The ball must leave the outer hoop, travel through the air, pass through the opening of the inner hoop and be captured by one of its rolling surfaces. Unlike T1, T2 and T3, which take place within a single rolling mode, T4 exploits the hybrid structure of this model.

6.6.1 Problem Statement

The flying ball manoeuvre consists of three successive phases. First, the ball rolls on the outer hoop and is accelerated until it loses contact with the track. It then enters free-flight in a ballistic trajectory, while the hoop is repositioned to bring the opening to the predicted impact location. Finally, the ball impacts the inner hoop and must be captured in a state from which it can be stabilised by the regulator developed for T1.

The problem can be formulated as a three-phase hybrid optimal-control problem since the mode sequence is known in advance. The durations of the phases and the continuous state trajectories are unknown, while the transitions are defined by guard conditions and reset maps.

Phase 1: Launch

During the first phase, the ball rolls on the outer hoop and is accelerated until it loses contact with the track. The dynamics are therefore the nonlinear rolling-mode dynamics used in T3, with the same actuator constraint. Unlike T3, however, contact is not required throughout the phase: liftoff is the objective. It occurs at the first instant at which the normal force vanishes,

$$N(\psi(t_{lo}), \dot{\psi}(t_{lo})) = 0. \quad (6.23)$$

The state at liftoff is not prescribed in advance. It belongs to the liftoff manifold defined by (6.23), and its position on this manifold determines the subsequent ballistic trajectory. The launch phase therefore cannot be designed independently of the phases that follow.

Phase 2: Free flight and alignment

Once in the air, the ball is mechanically disconnected from the hoop. Its trajectory is governed by the free-flight dynamics under gravity alone, and the control input no longer affects the ball. The hoop remains actuated and is used to position the opening at the predicted impact location.

If the ball reaches the inner hoop at time t_i and angular position ψ_i , alignment requires

$$|\theta(t_i) + \psi_{\text{hole}} - \psi_i|_{2\pi} \leq \frac{\Delta\psi_{\text{hole}}}{2}, \quad (6.24)$$

where $|\cdot|_{2\pi}$ denotes the smallest absolute angular difference modulo 2π . The flight duration is determined by the launch state and gravity, resulting in a rendezvous problem for the hoop.

Phase 3: Capture and stabilisation

The third phase begins when the ball reaches the inner-hoop surface. The impact is represented by the reset map introduced in the mathematical model (see [section 3](#)). This map determines the post-impact velocity and therefore the initial condition of the following rolling motion.

Geometric contact is not sufficient for successful capture. The post-impact state must remain within the region of attraction of the regulator designed in T1 for the newly entered rolling mode. The objective of the third phase is therefore to bring the ball to the bottom equilibrium after impact, while respecting the dynamics of the contacted surface.

Final formulation

The three phases are coupled through their transition conditions. The state is continuous at liftoff, whereas the impact introduces the reset map. The phase durations are decision variables, and the final impact must satisfy both the contact condition and the alignment requirement. The resulting problem is written as

$$\mathcal{P}_4 : \min_{u(\cdot), t_{lo}, t_i} J_4 = \int_0^{t_{lo}} u(t)^2 dt + \int_{t_{lo}}^{t_i} u(t)^2 dt + \Phi_4(x(t_i^+)), \quad (6.25)$$

subject to

- the dynamics corresponding to the current phase,
- the initial condition at the bottom of the outer hoop,
- the liftoff guard (6.23), state continuity at liftoff, and the impact guard and reset map at capture,
- the alignment condition (6.24),
- the actuator and contact constraints in the rolling phases.

No discrete mode-selection is required since the mode sequence is known in advance. The optimisation problem determines the phase durations, the continuous trajectories, and the control inputs, while enforcing the guards and reset map at the phase transitions. The post-impact stabilisation is not optimised directly within $\mathcal{P}_4$. It is handled by the T1 regulator after the reset. As in T3, the solution of $\mathcal{P}_4$ is a nominal trajectory. The next subsection describes its generation.

6.6.2 Trajectory Generation

The first temptation is to design the three phases sequentially: first choose a launch trajectory, then predict the ballistic flight, and finally check whether the hoop can be repositioned to meet the ball. This approach is not suitable for T4. The launch state determines the entire free-flight trajectory, while the resulting impact time and angle determine the motion required from the hoop. The launch must therefore be selected with the later alignment requirement already taken into account.

For a given liftoff state, the free-flight dynamics determine the impact time t_i and angular position ψ_i of the ball. During this interval, the hoop is described by the double-integrator dynamics $\ddot{\theta} = u$, with $|u| \leq u_{\max}$. If the hoop state at liftoff is $(\theta_{lo}, \dot{\theta}_{lo})$, the set of hoop angles reachable at the impact time is

$$\theta(t_i) \in \left[\theta_{lo} + \dot{\theta}_{lo} T_f - \frac{1}{2} u_{\max} T_f^2, \theta_{lo} + \dot{\theta}_{lo} T_f + \frac{1}{2} u_{\max} T_f^2 \right], \quad (6.26)$$

where $T_f = t_i - t_{lo}$ is the flight duration. Alignment is feasible only if this interval intersects the angular acceptance interval defined by (6.24), modulo 2π .

The trajectory is therefore generated by solving the three phases as one multi-phase problem. Each phase has its own continuous dynamics, duration, state trajectory and control input, and they are linked by the liftoff and impact guards and reset maps. The same Hermite-Simpson transcription is applied to each phase for this task, with the phase durations left as decision variables and the different conditions added as equality or inequality constraints in the resulting nonlinear program.

Its solution provides a nominal launch trajectory in phase 1, a ballistic trajectory for the ball and a repositioning trajectory for the hoop in phase 2, and a post-impact stabilisation in phase 3. The resulting trajectory is shown in [Figure 6.8](#). The three phases are shown together, with the mode transitions marked as for the open-loop scenarios (see [subsection 3.5](#)). The figure also shows the predicted position of the opening when crossing the inner hoop.

Solving the launch phase independently would select a liftoff state without considering the downstream alignment constraint. If the resulting impact lies outside the reachable interval of the hoop, the launch would have to be redesigned from scratch. The multi-phase formulation precisely avoids this.

6.6.3 Control Law

The control architecture switches between three phase-dependent laws, according to the active mode. During the launch phase, the TVLQR feedback developed for T3 is used to track the nominal rolling trajectory.

$$u(t) = u^*(t) - K(t)(x(t) - x^*(t)). \quad (6.27)$$

During free flight, the hoop is commanded to bring the opening to the predicted position of the opening at impact. After capture, control switches to the T1 regulator associated with the newly entered rolling mode,

$$u(t) = u_{eq} - K(x(t) - x_{eq}). \quad (6.28)$$

The switches are triggered by detected guard events, which makes the process robust to tracking errors that could shift the actual liftoff and impact times.

6.6.4 Results

The success of the whole manoeuvre depends on the consistency of the transitions, and not only on the performance of each controller within its own phase. The main uncertainty lies at the transitions. Liftoff detection is affected by state-estimation delay, while the capture essentially depends on the validity of the reset map after landing. The reset map is a modelling assumption that has not been validated on the prototype. Its influence is one of the first things to check when working on the prototype.

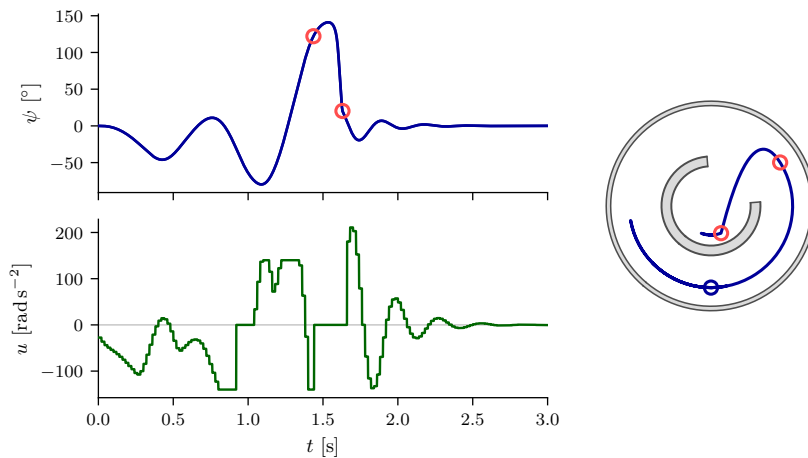


Figure 6.8: Multi-phase trajectory of the flying ball manoeuvre, with the transitions marked in red.

In simulation, the last task works. Like for T3, it does not guarantee that the controller works on the real plant until the considerations discussed for T3 are solved.

7 Conclusion

This thesis addressed the design, modelling, identification and control of a flying ball-in-a-hoop apparatus intended for the teaching of control engineering.

A hybrid model of the double-hoop system was established. It covers the three modes described in the literature and adds a fourth one, made possible by the opening cut in the inner hoop, together with the guards and reset maps connecting them. A prototype was then designed from the requirements of the four tasks, built and identified. The assembly departs from the idealised model on one point that turned out to matter: the ball is carried by two O-rings rather than by a single contact, which raises its effective rotational inertia by a factor 1.552 instead of 1.4. The two-rail model predicts a natural frequency within 1 % of the measured 1.311 Hz, against 5 % for the idealised one. Validated against a trajectory it had not been fitted to, the model gives a root mean square error of 2.47° on the ball position.

Two of the four tasks were achieved on the prototype. Stabilisation and reference tracking were obtained with a linear quadratic regulator, and two limitations were measured rather than assumed. With identical gains, the closed loop is stable at 50 Hz and divergent at 36 Hz, which locates the instability in the latency of the vision chain rather than in the choice of weights. And the time constant of the inner velocity loop, which falls from 196 ms to 17 ms once its gains are recomputed, governs the tracking performance more than the outer design does.

The loop-the-loop was planned by direct collocation and stabilised by a time-varying regulator. It succeeds in simulation and fails on the bench at approximately 98° , reproducibly. The cause is a limitation of the model rather than a defect of the controller. Imposing rolling without slipping removes the tangential contact force from the equations, so the optimiser has no variable through which the law of Coulomb could be violated. Reconstructed a posteriori, the nominal trajectory demands $\mu_{req} \approx 3.7$, far above what the contact can supply, and raising the gains only raises that requirement further. The flying ball task was therefore studied in simulation only, its launch phase resting on the same model.

Three improvements follow. The tangential contact force should be kept in the planning model, with the law of Coulomb imposed as a path constraint. The impact reset map, which absorbs most of the energy carried into the impact and therefore cannot discriminate a ball caught from a ball that would have bounced away, should be identified from measurements instead of assumed. And the effect of the sensing delay should then be studied again on the corrected model. The apparatus offers a real progression of experiments, from linearisation and quadratic regulation to trajectory optimisation, hybrid dynamics and contact modelling, which is what a didactic device should provide. The parametric CAD models, the simulation code and the software running on the bench are provided alongside this manuscript.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the author used Claude in order to generate short Python scripts for the visualisation of the data recorded on the Raspberry Pi, and to assist with the drafting and revision of the manuscript.

After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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A Complete friction model for the ball

The model of [section 3](#) presents all dissipation acting on the ball by a single viscous coefficient b_b . This appendix explains the different mechanisms that were considered, and justifies this choice. Four different mechanisms are considered: static friction that enforces rolling, viscous dissipation, rolling resistance, and aerodynamic drag. Each is estimated from the identified geometry of [section 4](#). All estimates are made using the configuration of experiment E2 (see [subsection 4.4](#)): the hoops are held fixed, and the ball oscillates freely on the o-ring tracks of the outer hoop. Under no slip assumption, the system reduces to a single degree of freedom ψ , and the equation gives:

$$\underbrace{k m_b R_{eff}^2}_{J_\psi} \ddot{\psi} + \tau_{diss}(\psi, \dot{\psi}) + m_b g R_{eff} \sin \psi = 0, \quad k = 1 + \frac{2}{5} \left(\frac{r_b}{r_R} \right)^2, \quad (\text{A.1})$$

where R_{eff} is the radius of the circle described by the ball centre and r_R the rolling radius at the two-rail contact (see [section 5](#)), $k = 1.552$, $m_b = 67$ g, $R_{eff} = 91.6$ mm, $r_b = 12.5$ mm, $r_R = 10.64$ mm. Linearising this equation gives $\omega_0 = \sqrt{g/(k R_{eff})} = 8.26$ rad/s, leading to $f_0 = 1.3144$ Hz.

The ball is supported by two o-rings rather than by a single contact point. The two normal forces are therefore inclined at α to the radial direction (here, $\alpha = 31.7^\circ$). Their sum therefore exceeds the radial load by $1/\cos \alpha = 1.18$, leading to:

$$N = \frac{1}{\cos \alpha} \left(m_b g \cos \psi + m_b R_{eff} \dot{\psi}^2 \right). \quad (\text{A.2})$$

Coulomb sliding friction This term does not appear in the model because the contact is assumed to roll without slipping. This assumption can be verified by projecting the ball equation onto the tangential direction. The static friction force required to enforce rolling is then

$$F_f = m_b g \sin \psi \frac{k-1}{k}, \quad (\text{A.3})$$

which reduces to $\frac{2}{7} m_b g \sin \psi$ of a solid sphere on a flat surface, when $r_R = r_b$. Dividing this by [\(A.2\)](#) gives the required static friction coefficient:

$$\mu_{req}(\psi) = \frac{k-1}{k} \cos \alpha \tan \psi = 0.303 \tan \psi. \quad (\text{A.4})$$

ψ [deg]	10	20	30	45	55
μ_{req} [-]	0.053	0.110	0.175	0.303	0.432

Table A.1: Static friction coefficient required to maintain rolling contact, from [\(A.4\)](#).

For a steel-NBR pair, $\mu_s \approx 0.5$ -1.0. Rolling is therefore ensured before any sliding occurs, until $\psi \approx 60^\circ$.

Viscous dissipation The viscous model is used in the equations of motion through a Rayleigh dissipation function $D = \frac{1}{2} b_b \dot{\phi}^2$, where ϕ is the spin angle of the ball. With the hoop blocked, the no-slip condition gives $b_b (R_{eff}/r_R)^2 \dot{\psi}$. The generalised damping torque is therefore $b_b \left(\frac{R_{eff}}{r_R} \right)^2 \dot{\psi}$, and

inserting it into the equation of motion gives

$$\ddot{\psi} + \frac{b_b}{km_b r_R^2} \dot{\psi} + \omega_0^2 \psi = 0 \quad (\text{A.5})$$

An important point is worth noting: the rolling radius r_R enters (squared) in the viscous term. During experiments, the value of b_b is deduced from the decay rate of the ball motion. Therefore, an error in the contact geometry propagates quadratically into b_b , which is why the two-rail correction cannot be skipped.

Aerodynamic drag Even though it is supposed to be negligible, aerodynamic drag can be derived. At the Reynolds numbers reached here, the drag on a sphere is quadratic in velocity.

$$F_D = \frac{1}{2} C_D \rho_{air} \pi r_b^2 v^2, \quad \text{Re} = \frac{\rho_{air} v 2r_b}{\mu_{air}}. \quad (\text{A.6})$$

where $\rho_{air} = 1.2 \text{ kg/m}^3$, $\mu_{air} = 1.81 \cdot 10^{-5} \text{ Pa} \cdot \text{s}$ and $C_D \approx 0.47$ for a smooth sphere.

The largest velocity encountered on this apparatus is the one required at the top of the hoop during the looping manoeuvre. Evaluated at that point, the drag force remains below 10^{-4} , against contributions higher by several orders of magnitude.

In contrast, other contributions are higher by several orders of magnitude. Aerodynamic drag accounts for less than 0.01% of the energy budget. This contribution is therefore discarded without further discussion.

Rolling resistance When a ball rolls on a deformable surface, the normal pressure distribution in the contact patch is asymmetric: material ahead of the contact is compressed, while material behind the ball is recovering. The recovery returns less energy than the compression, leading to a resultant force producing a resisting torque on the contact point. [28].

In other words, a force $F_r = \mu_r N$ opposes the motion of the ball at its centre, with μ_r the (dimensionless) coefficient of rolling resistance. The most important property of this force is that it depends on the normal load but not on the velocity itself. It only changes once the motion changes of direction.

$$\tau_c = \mu_r N R_{eff} \text{sgn}(\dot{\psi}) \quad (\text{A.7})$$

The amplitude of this contribution is deduced during the identification experiments (see [section 5](#)). Its amplitude is clearly not negligible. It is however left out of the model, for two main reasons. Firstly, all tasks are built on a first-order expansion of the dynamics about an equilibrium or a nominal trajectory. The derivative of $\text{sgn}(\dot{\psi})$ with respect to $\dot{\psi}$ is a Dirac distribution at $\dot{\psi} = 0$. No finite-state matrix A can contain this term. Secondly, the neglected torque always opposes the motion, so the power it delivers is negative for all $\dot{\psi}$ (passive perturbation). Adding this term to V , a Lyapunov function establishing the stability of the closed loop for the model, can only make $\dot{V}$ more negative than it is [43].

B Field-Oriented Control

The dynamic model treats the actuator as a pure source of torque (see [section 3](#)). The hoop torque τ enters the equations of motion directly, without any electrical consideration. This assumption is not evident, since the motor is a three-phase permanent magnet machine, whose torque depends on the rotor angle, phase currents and their time derivatives. Field-Oriented Control(FOC) allows to reduce this to a single scalar gain $\tau = K_t i_q$.

In a brushed DC motor, the mechanical commutator switches the winding currents as the motor turns, so that the current carrying conductors always stay in the same position relative to the magnetic field. The torque is therefore proportional to the terminal current regardless of the rotor angle, and the machine can be described by a single constant. In a brushless machine, there are no commutators. The three stator windings are fixed and their currents must be switched electronically. The contribution of each phase to the total torque varies with the electrical rotor angle ($\theta_e = p \theta$, with p the number of poles of the machine, 7 for the D5065 and θ the mechanical angle measured by the encoder). Commanding a fixed current therefore produces a torque that varies with the rotor position.

FOC helps to solve the problem. It continuously reshapes the three phase currents so that the resulting stator field is held at a constant angle to the rotor field. The machine then behaves exactly like a brushed DC motor, from the outside point of view.

Clarke-Park transformations The three phase currents are not independent. In a star-connected winding with no neutral wire, the current law impose $i_a + i_b + i_c = 0$, so only two degrees of freedom are left on the measured currents. The Clarke transformation makes this explicit, by projecting them on an orthogonal stator fixed frame (α, β) [\[44\]](#).

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}. \quad (\text{B.1})$$

This transformation replaces the three coplanar axes at 120° by two perpendicular ones, that still rotate at the electrical frequency. It is therefore still time-varying (even though these sinusoidal quantities become constants when reaching the steady-state). The Park transformation removes the remaining time dependence by expressing the current vector in a frame, called dq frame, that rotates with the rotor flux [\[44\]](#).

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \quad (\text{B.2})$$

The direct (d) axis is aligned with the rotor magnetic flux, while the quadrature (q) axis is 90° electrical in advance from it. The transformation requires θ_e , and therefore requires the rotor position at each control step (the use of an encoder is therefore a functional prerequisite for the actuator itself, and not only a feedback sensor for the outer loop).

Torque expression Applying the transformation to the three-phase voltage equations leads to

$$\begin{cases} v_d = R i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q, \\ v_q = R i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_m) \end{cases} \quad (\text{B.3})$$

with R the phase resistance, L_d, L_q the axis inductances, λ_m the permanent-magnet flux linkage and $\omega_e = p\dot{\theta}$. Each axis is a first-order electrical circuit, coupled to the other through speed dependent terms, and driven by the back-EMF $\omega_e \lambda_m$. The electromagnetic torque is

$$\tau = \frac{3}{2} p \left[\underbrace{\lambda_m i_q}_{\text{magnet}} + \underbrace{(L_d - L_q) i_d i_q}_{\text{reluctance}} \right]. \quad (\text{B.4})$$

The D5065 motor is a surface-mounted permanent magnet machine. The magnets therefore sit on the rotor surface, and their relative permeability is close to that of air. The magnetic circuit therefore presents the same reluctance in every direction, hence $L_q \approx L_d$. The reluctance term vanishes, and

$$\tau = \frac{3}{2} p \lambda_m i_q = K_t i_q. \quad (\text{B.5})$$

Control structure The torque depends on i_q alone, while the current i_d only dissipates $R i_d^2$ in the windings. For a fixed current magnitude, $|i|^2 = i_d^2 + i_q^2$, the torque is therefore maximised when the entire current is put on the q axis. The controller commands $i_d^{ref} = 0$.

In practice, the control loop is executed by the ODrive S1 at the PWM rate. Each cycle performs several steps

- measure the two phase currents, the third following from $i_a + i_b + i_c = 0$, and the encoder position θ ,
- apply the transformation to obtain i_d and i_q ,
- compute the errors $i_d^{ref} - i_d$ and $i_q^{ref} - i_q$, and pass them through PI regulators to produce v_d and v_q ,
- add the feedforward decoupling terms to make sure that each axis behaves as an independent first-order system,
- apply the inverse transformation to generate the three currents.

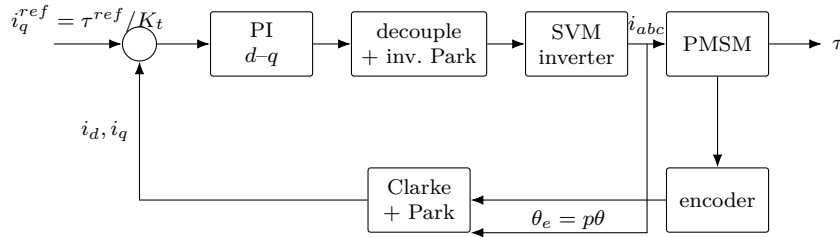


Figure B.1: Field-oriented current control loop as implemented on the ODrive S1. The outer loop control laws enter at the left of this loop, as a torque request converted to i_q^{ref} through the constant K_t .

C Inner Control Loop

The actuator delivers a torque proportional to the commanded current (see [Appendix B](#)). The model actually used by the control laws is however a bit more complex: the interface uses a velocity regulator (in the driver directly). This section aims to establish and validate the assumption $\ddot{\theta} = u$, on which all the controllers depend.

Velocity-mode law Seen from the shaft, the rotating assembly (rotor, hoops and supports) is a single inertia subject to bearing friction. The equation of motion resulting from this point of view is

$$J\ddot{\theta} + b\dot{\theta} = \tau, \quad P(s) = \frac{\dot{\Theta}(s)}{T(s)} = \frac{1}{Js + b} = \frac{1}{b} \frac{1}{\omega_b s + 1}. \quad (\text{C.1})$$

This equation corresponds to a first-order low-pass with a static gain of $1/b$ and a corner frequency at $\omega_b = b/J$. The two parameters of this equation are the one identified by coast-down in experiment E1b (see [section 5](#)): $J = 1.1193 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2$, $b \approx 6 \cdot 10^{-5} \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad}$. This places the corner at $\omega_b = 5.5 \cdot 10^{-2} \text{ rad/s}$, or $9 \times 10^{-3} \text{ Hz}$. The ball resonates at 1.31 Hz , which is four orders of magnitude above it, and the control loop runs at 50 Hz . The pole resulting from the motor friction is therefore completely invisible over the entire useful band, and the transfer function of the motor can reduce to $P(s) \simeq \frac{1}{Js}$. This corresponds to a pure integrator, with $-20\text{dB}/\text{decade}$ of magnitude, and a constant -90° of phase. The open-loop machine has no natural frequency regulation, which is why the driver closes a loop around it.

The experiments of this work use the machine in velocity-mode. The control law produces an acceleration, u , which is integrated into a velocity setpoint. This setpoint is sent to the driver, which regulates the motor around it using its own PI controller, acting on the velocity error.

$$\tau^{ref} = K_p (\dot{\theta}^{ref} - \dot{\theta}) + K_i \int (\dot{\theta}^{ref} - \dot{\theta}) dt. \quad (\text{C.2})$$

The units of all angular quantities are expressed in turns, and not in radians in the driver. The two configuration parameters, `vel_gain` in $\text{N} \cdot \text{m}$ per turn/s and `vel_integrator_gain` in $\text{N} \cdot \text{m}$ per turn, relate to the gains of [\(C.2\)](#) by $K_p [\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}] = \text{vel_gain} / 2\pi$ and $K_i [\text{N} \cdot \text{m}/\text{rad}] = \text{vel_integrator_gain} / 2\pi$.

Closed-loop transfer function Applying the PI controller to the equation of the machine leads to the closed-loop transfer function of this loop.

$$G_v(s) = \frac{\dot{\Theta}(s)}{\dot{\Theta}^{ref}(s)} = \frac{K_p s + K_i}{J s^2 + K_p s + K_i}, \quad \omega_n = \sqrt{\frac{K_i}{J}}, \quad \zeta = \frac{K_p}{2\sqrt{K_i J}}. \quad (\text{C.3})$$

The numerator of this equation carries a zero at $-K_i/K_p$, which makes this transfer function impossible to read from its poles alone, but becomes a powerful tool when trying to design this loop: adapting the value of both gains to get a ratio that make this zero coincide with a pole is an easy way to cancel their contributions.

When designing the system with this objective in mind, only one fast pole survives,

$$G_v(s) \simeq \frac{1}{1 + \tau s}, \quad \tau = \frac{J}{K_p}. \quad (\text{C.4})$$

This first-order reduction is used through the work to interpret the different step responses but also to improve the simulation by taking the dynamics of the motor into account in a simple way. A simple parameter can be used to select whether the ideal assumption is used or the first order model to express the dynamics of the motor (`cfg.motor_model = 'first_order' or 'ideal'`).

Gains tuning

A velocity step is commanded and the response of the drive recorded, first with the factory gains and then after retuning (`vel_gain = 0.1667`, `vel_integrator_gain = 0.3333`). This set of gains leads to $K_p = 2.653 \cdot 10^{-2} \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ and $K_i = 5.305 \cdot 10^{-2} \text{ N}\cdot\text{m}/\text{rad}$.

The gains are recomputed so that the closed inner loop behaves as a first-order system of prescribed time constant, and the computed values are written to the driver. **Figure C.1** shows the three responses, and **Table C.1** collects their characteristics. The final retained values are `vel_gain = 1.0461` and `vel_integrator_gain = 0.10`, or $K_p = 1.665 \cdot 10^{-1} \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ and $K_i = 1.592 \cdot 10^{-2} \text{ N}\cdot\text{m}/\text{rad}$. This leads to a zero placed at $-K_i/K_p = 0.0956 \text{ rad/s}$, exactly where the slower pole leads. The cancellation on which the first order assumption rests becomes almost exact.

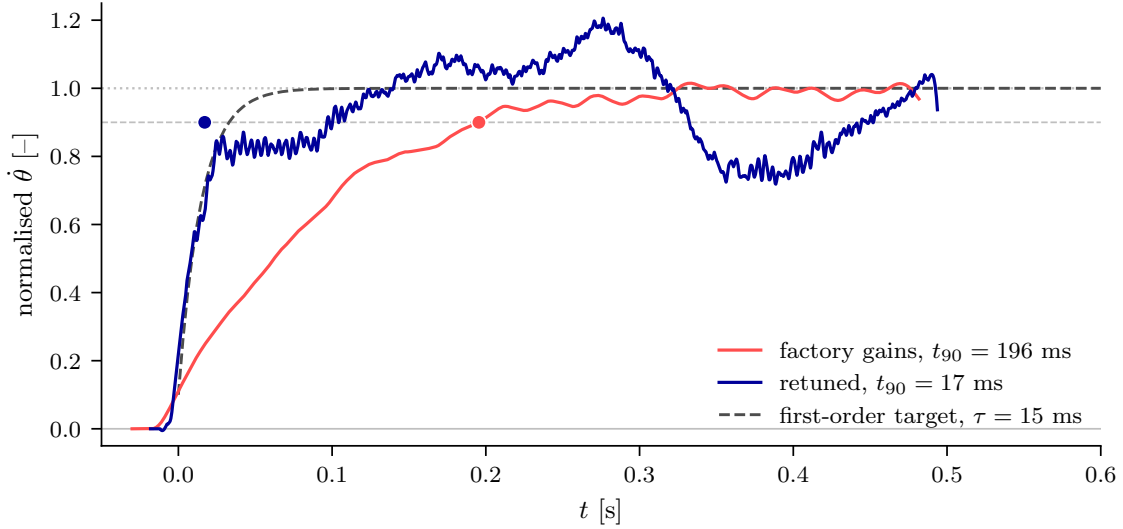


Figure C.1: Step response of the inner velocity loop, with the factory gains, with the first-order response targeted by the design, and with the computed gains applied to the bench.

Configuration	t_r [ms]	t_s [ms]	M_p [%]
Factory gains	195.5	282.5	1.6
Retuned bench	12.4	17	39.0

Table C.1: Rise time, settling time and overshoot of the three step responses of **Figure C.1**.

D Camera model and Image Acquisition

The measurement of ψ rests on two properties stated when the sensor was selected (see [subsubsection 4.3.3](#)): the recovered angle does not depend on the focal length nor on the working distance, and its precision follows from the number of pixels the apparatus occupies. Both are established here, together with the sensitivity of the measurement to the alignment the support is built to enforce.

Projection model The sensor is described by the pinhole model [\[45\]](#). A point (X, Y, Z) expressed in the camera frame projects onto the image plane at

$$u = f_u \frac{X}{Z} + c_x, \quad v = f_v \frac{Y}{Z} + c_y, \quad (\text{D.1})$$

where f_u and f_v are the focal lengths in pixels along the two image axes and (c_x, c_y) is the projection of the optical axis, the reference centre recovered during commissioning (see [subsection 5.2](#)).

The camera support constrains the optical axis to be perpendicular to the plane of the hoops and to pass through the rotation axis. Under these two conditions, a ball rolling at the effective radius R_{eff} has coordinates

$$(X, Y, Z) = (R_{eff} \cos \psi, R_{eff} \sin \psi, d_{cam}), \quad (\text{D.2})$$

with d_{cam} constant throughout the motion, since the ball stays in a plane parallel to the image plane. Substituting into [\(D.1\)](#) and taking the offsets from the reference centre,

$$\Delta u = u_b - c_x = f_u \frac{R_{eff}}{d_{cam}} \cos \psi, \quad \Delta v = v_b - c_y = f_v \frac{R_{eff}}{d_{cam}} \sin \psi. \quad (\text{D.3})$$

For square pixels, $f_u = f_v$, and the factor $f R_{eff} / d_{cam}$ is common to both offsets. It therefore cancels in their ratio, and

$$\psi_{meas} = \text{atan2}(\Delta v, \Delta u). \quad (\text{D.4})$$

Three quantities disappear in this expression, and it is worth being explicit about which. The focal length cancels, so no calibration of f is required. The working distance cancels, so the rail may be moved without recalibrating the angle. The effective radius cancels as well, which means the same relation holds on all three rolling surfaces and during free flight, where r is no longer constant. What does not cancel is (c_x, c_y) : an error on the reference centre biases ψ directly, which is why it is measured on the assembled apparatus rather than taken as the centre of the image.

Propagation of the centroid uncertainty The centroid returned by the segmentation carries an uncertainty σ_{px} , which [\(D.4\)](#) converts into an uncertainty on the angle. Writing $\rho = \sqrt{\Delta u^2 + \Delta v^2}$ for the pixel distance from the reference centre, the partial derivatives of the arc tangent give

$$\sigma_\psi = \frac{\sigma_{px}}{\rho} = \frac{\sigma_{px} d_{cam}}{f R_{eff}}. \quad (\text{D.5})$$

The uncertainty is inversely proportional to the radius of the circle described by the ball in the image, so a given pixel error costs more angle on the inner hoop than on the outer one, in the ratio of their effective radii. Taken with the dispersion measured on the reference centre, [\(D.5\)](#) is what places the uncertainty of the vision chain below the target of 1° set during the design.

Two consequences follow for the placement of the camera. Bringing it closer raises ρ and lowers σ_ψ , which argues for the shortest working distance the rail allows. Against this, the rim of the outer hoop must remain inside the field of view with enough margin for the Hough transform to find it, which

sets the opposite bound. The travel of the rail was dimensioned to span exactly this trade-off, and the retained distance sits inside it rather than at either extreme.

Sensitivity to misalignment The cancellation above assumes the optical axis to be perpendicular to the plane of the hoops. If the camera is tilted by a small angle ε about an axis lying in the image plane, the circle described by the ball projects as an ellipse rather than a circle, of axis ratio $\cos \varepsilon$. The recovered angle then carries a harmonic error

$$\delta\psi(\psi) \simeq \frac{1}{2} \varepsilon^2 \sin(2\psi), \quad (\text{D.6})$$

which vanishes at the four cardinal positions and is largest halfway between them. Two features of this error matter. It is quadratic in the tilt, so the perpendicularity need only be enforced to a few degrees for it to stay well below the pixel-level uncertainty of (D.5). But it is deterministic and periodic in ψ , so it is not reduced by averaging over trials, and it would appear in an identification campaign as an apparent dependence of the dynamics on the ball position. This is the reason the support removes every degree of freedom except the working distance rather than relying on an adjustment made once. A lateral offset of the camera, by contrast, displaces the projection of the rotation axis without distorting the angles, and is absorbed entirely by (c_x, c_y) . This is what makes the centring of the apparatus in the frame a matter of convenience rather than of accuracy, provided the reference centre is measured.

Acquisition chain and its timing The segmentation follows the architecture of the reference implementation [21], whose parameters were set on the present apparatus during commissioning. Its structure is a coarse search over the full frame followed by a fine search restricted to a window centred on the previous detection. The two stages exist for different reasons: the coarse search recovers the ball after a loss of detection and costs a full-frame pass, while the tracking window bounds the cost of a nominal frame and is what allows the chain to fit inside the control period.

The width of that window is the one parameter of the chain that couples the two ends of the trade-off. It must exceed the displacement of the ball between two frames, $\Delta_{px} \simeq \lambda^{-1} R_{eff} \dot{\psi} T_s$, or the ball leaves the window and the coarse search is triggered, costing a full frame at the instant the ball moves fastest. It must also stay small enough for the processing to complete within T_s . The value retained during the identification campaign satisfies both at the velocities encountered there, and is the quantity to revisit first if a faster task shows an irregular sampling period.

The total delay of the chain, measured on the assembled bench rather than taken from a datasheet, is close to one control period. Its consequence on the closed loop is not a modelling detail: it is the phase it consumes that bounds the loop rate at which the regulator of subsection 6.3 remains stable, and it enters the delay budget.